

Data Complexity, Part II: what the encoder removes before the model is reached

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Part I proposed that *data complexity* governs when quantum machine learning offers an advantage, and that a composite complexity measure could predict the achievable quantum edge. Under pre-specification, this hypothesis failed: pooled leave-one-dataset-out $R^2 = -0.32$, below the pre-specified threshold of 0.4. Here we analyze why, and find that the failure is structural rather than merely statistical.

The measured edge depends strongly on the classical baseline. Against logistic regression, the selector predicts held-out performance ($R^2 = +0.13$), with the parity family winning 17% of cells. Against the best of six standard classical models, performance falls to $R^2 = -0.94$, with parity winning only 1%. The apparent signal therefore tracks where a linear model fails, rather than an intrinsic quantum advantage.

The encoding procedure further removes the structure the selector is asked to predict. Reduction to few-qubit width leaves little nonlinear headroom, while gradient boosting matches a linear probe at every width. Moreover, the reduction is not free to implement coherently: feature selection reaches the exact information floor $(p - d)b$ with zero ancillas, whereas an orthonormal rotation requires rounding and 84 ancillas on a 32-qubit register. Because an encoder acting on classical data is itself a classical computation, it cannot raise the quantum ceiling; it can only raise the classical floor.

Finally, the regime itself limits what any predictor can rank: the quantum model is at chance in 81.5% of cells. Yet on data constructed to carry a quantum signature, a +0.42 edge emerges under a designed measurement, while models required to learn the structure remain at chance. These results show that data complexity alone cannot predict quantum advantage: the observable edge is jointly determined by the classical baseline, the encoding-induced loss of headroom, and whether the quantum structure is present and accessible to the measurement.

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1 Introduction

Debate over when quantum machine learning can outperform classical methods has been conducted largely in terms of the quantum side of the pairing: the circuit family, the encoding, the ansatz, and the hardware. Part I (Pere, 2025) argued that this leaves out the variable that should determine the answer, and proposed *data complexity* as the missing axis: the structural, statistical, algorithmic, and topological richness of the dataset itself. It formalized that intuition as four pillars, entropy, correlation order, compressibility, and topology, combined into a normalized composite $\tilde{C}_{\text{data}}$, and used the composite to make three quantitative predictions. Barren-plateau onset should scale as $\text{Var}[g(\theta)] \approx \exp(-\alpha \tilde{C}_{\text{data}})$, so that gradients vanish faster on more complex data. Generalization error should be governed by the *alignment* between a circuit’s expressibility and the data’s complexity, penalizing mismatch in either direction. And, integrating both, the achievable quantum edge should be *ranked* by $\tilde{C}_{\text{data}}$, making the composite a practical instrument for deciding, before running anything, whether a given dataset is worth a quantum treatment at all. That last claim is what would make the framework useful rather than merely descriptive, and it is the one this paper sets out to test.

Part II is that test. Before testing the empirical predictions, however, preparing the experiments exposed problems in Part I’s formalism: dimensionally

incoherent terms in the composite, correlated components that make its weights unidentifiable, and a data-free limit that contradicts the circuit-ensemble result of McClean et al. (2018). These issues were corrected before the confirmatory analysis was run (Sec. 3). We treat this as scientific iteration rather than a change to the hypothesis: the core proposition, that data complexity is the relevant axis, is preserved throughout; what changes is how it is operationalized, and then whether it survives contact with data.

A failed prediction is informative only if the experiment can distinguish failure of the hypothesis from failure of the measurement mechanism. We therefore fixed a falsification battery with explicit decision rules before collecting confirmatory data. More importantly, we can attribute the measured quantum–classical gap to the part of the pipeline that actually produces it. Any measured gap decomposes exhaustively into a term obtainable by running the preprocessing classically and a residual that is the only quantum claim (Sec. 6.3). **This decomposition changes how we should interpret the negative result: rather than asking only whether the complexity predictor succeeds, we can ask whether any quantum quantity remains for it to predict.**

The analysis yields four lessons.

1. **The baseline determines the measured advantage** (Sec. 6.2). A fully predictable “advantage” appears and disappears depending on the classical model set alone. Against a linear baseline, the selector recovers a signal; against the best of six standard models, that signal largely vanishes.
2. **The encoding removes the headroom it is meant to predict** (Sec. 6.3). Reducing classical data to few-qubit width removes nonlinear headroom before the quantum model sees it. Gradient boosting matches a linear probe at every width, and where a baseline-robust edge survives, gradient-based training cannot find it. The encoding itself is a classical computation: it cannot raise the quantum ceiling, only the classical floor.
3. **The scaling law is misspecified where experiments live** (Sec. 6.4). Varying circuit width shows power-law, depth-independent decay rather than the exponential form assumed by the framework, and the observed scaling is inconsistent with the Haar-random rate.
4. **The test had nothing to detect, and that bounds the null** (Sec. 6.5). The quantum model sits at chance in 81.5% of cells, so the edge degenerates to classical error ($\rho = +0.913$) and the encoding axis carries only 0.5% of the variance. Where there is no varying quantum signal, no framework can rank it. There is no quantum signal left to extract beyond what a classical model can find.

Hypotheses, decision rules, the complexity and interaction vectors, the predictor architecture, the encoding library, the cost model, and the cross-validation scheme were fixed in a written pre-specification before any analysis was run. Each result is therefore a measurement of one of the two terms in the attribution: Sec. 6.2 shows how much of an apparent edge the classical term accounts for; Sec. 6.3 shows what the encoding destroys and costs before a quantum residual can exist; Sec. 6.4 tests whether the proposed scaling law describes the regime in which the experiments operate; and Sec. 6.5 identifies the regime in which the quantum term is pinned and therefore unmeasurable.

The question forced by this decomposition is therefore more fundamental than whether one composite score predicts the edge. *Under what experimental conditions is there a genuinely quantum quantity left to predict?* That is the question this paper leaves behind.

Main takeaways:

- **Baseline dependence:** the apparent quantum edge is strongly baseline-dependent: ($R^2 = +0.13$) against LR, but (-0.94) against the best of six classical models.
- **Attribution:** the measured edge is largely explained by classical baseline error rather than an encoding/data-complexity signal.
- **Variance/headroom:** the quantum model is at chance in 81.5% of cells, while the encoder reduces the nonlinear headroom available to the quantum model.
- **Control:** a designed quantum-signature dataset produces a ($+0.42$) edge under structure-aware measurement, while generic models remain at chance.
- **Benchmark and generalization scope.** The best-of-six result is a fixed-library sensitivity analysis rather than a tuned classical frontier, and the 27-dataset leave-one-dataset-out test supports conclusions within the tested regime rather than universal generalization across QML tasks.

These results shift the question from whether a scalar data-complexity measure predicts quantum advantage to whether the relevant quantum structure survives the classical encoding and remains accessible to the measurement.

2 Related work

Data as a constraint on advantage. Whether a quantum model can help on a learning task is con-

strained by the data, not only by the circuit. [Huang et al. \(2021\)](#), for example, derive projected-kernel bounds that can rule out advantage under specified conditions and provide a prescreening criterion for when advantage may be possible from properties of the learning problem, while [Caro et al. \(2022\)](#) relate generalization from few training samples to circuit structure. These results motivate treating the dataset as an axis of quantum advantage, but they do not by themselves specify which properties of the data should be predictive in an experimental comparison. Part I proposed *complexity*, entropy, correlation order, compressibility, and topology as the operative axis and combined them into a linear composite. Part II instead replaces the fixed composite with a learned predictor and tests whether data complexity can predict the measured quantum–classical edge. In this sense, Part II revises the proposed operationalization rather than extending the framework.

The classical comparator sets the observable edge.

A quantum–classical comparison does not measure a property of the quantum model alone: it measures a difference between two model classes. Recent benchmark studies consistently find that tuned classical baselines remain competitive when comparisons are made carefully ([Bowles et al., 2024](#); [Yu et al., 2026](#); [Leither et al., 2026](#)), while end-to-end accounting can further change the apparent advantage ([Chandarana et al., 2026](#); [Gillani et al., 2026](#)). This makes the choice of classical model part of the quantity being measured rather than a secondary implementation detail. Our analysis therefore treats baseline choice as a first-order experimental variable. The resulting question is not simply whether a quantum model outperforms a classical one, but whether the observed edge survives a stronger classical comparator.

Trainability constrains what advantage can be observed.

Even when a quantum model has a representational advantage, that advantage is observable only if the relevant structure can be reached by the training procedure. Barren plateaus ([McClean et al., 2018](#); [Larocca et al., 2025](#)), cost-function locality ([Cerezo et al., 2021](#)), and kernel concentration ([Thanasilp et al., 2024](#)) characterize regimes in which variational or kernel-based quantum models become difficult to train or distinguish. H1 and H2 therefore test not only whether the scaling proposed in Part I is observed, but whether its functional form describes the regime in which the experiments actually operate. [Alcántara et al. \(2026\)](#) provide exact cost-variance expressions for arbitrary states and observables, removing the dynamical-Lie-algebra membership assumption that excluded settings such as ours; Sec. 6.4 compares the observed scaling with this broader framework. Conversely, [Cerezo et al. \(2025\)](#) emphasize that the absence of a barren plateau can it-

self be evidence for classical simulability. Trainability is therefore not independent of advantage: a model that is easy to train may also be one whose behaviour admits an efficient classical description.

Encoding changes the problem before the quantum model sees it. Data must generally be transformed before a few-qubit quantum model can process it, making the encoding inseparable from the comparison. Prior work shows that the embedding choice can dominate performance in few-qubit settings (Sammartino, 2026; Kurt, 2026); H4 asks whether this choice can be optimized at the feature level. We instead examine the step that precedes the encoder. If dimensionality reduction or feature selection is performed classically, it can remove precisely the nonlinear or high-order structure that a quantum model might otherwise exploit. Sec. 6.3 therefore asks what information and nonlinear headroom are lost in the reduction, what resources are required to reproduce it coherently, and how much of the eventual quantum–classical gap can be attributed to this classical term before the quantum model is run.

Data access and dequantization. A quantum advantage is also conditional on how the classical and quantum algorithms access the data. This point is central to the dequantization literature. Tang (Tang, 2019; Jethwani et al., 2020) showed that the quantum recommendation algorithm of Kerenidis and Prakash (Kerenidis and Prakash, 2016) could be matched, up to polynomial factors, by a classical algorithm under an ℓ_2 -sampling data structure, removing the claimed exponential speedup under that input model. Subsequent work generalized this idea to a broader sampling-based framework for low-rank matrix arithmetic and quantum singular value transformation, yielding quantum-inspired classical algorithms for recommendation, principal-component analysis, clustering, support-vector machines, regression, and related tasks (Chia et al., 2020). These results do not imply that quantum speedups are impossible; rather, they show that the input model is part of the speedup claim. A quantum algorithm can only claim an advantage over classical alternatives that are denied no more information or access than the quantum procedure itself. H6 therefore stratifies our benchmark by data-access mode. The present suite contains no Mode-Q pairs, so H6 is a descriptive boundary condition rather than a statistical test. We report this explicitly rather than treating the absence of qualifying instances as evidence for or against dequantization.

Taken together, these considerations place the data-complexity hypothesis inside a broader measurement problem. Data may determine where a quantum advantage is possible, but the *measured* edge also depends on what the classical comparator can learn, what structure survives preprocessing and encoding,

whether the quantum model can reach the relevant structure, and how the underlying data are accessed. These constraints motivate the attribution framework tested in the remainder of the paper: before asking whether data complexity predicts a quantum edge, we first identify how much of that edge can exist after the classical parts of the pipeline have been accounted for.

3 Update Framework

This part will provide some important improvement and updates from Part I.

3.1 From a scalar complexity score to an attributable framework

Part I treated data complexity as a scalar quantity, C_{data} , obtained by combining four pillars: entropy, correlation order, compressibility, and topology. Before testing whether this quantity predicts a quantum edge, Part II first asks whether the four pillars can legitimately be combined in this way. Two problems arise: the components are not commensurate on their native scales, and their statistical dependence makes the weights of a linear combination non-identifiable.

Equation 7 of Part I expressed the composite classical complexity as a linear sum,

$$C_{\text{data}} = \lambda_1 S(D) + \lambda_2 I_{\text{corr}}(D) + \lambda_3 K(D) + \lambda_4 C_{\text{top}}(D), \quad (1)$$

with weights $\{\lambda_i\}$ calibrated on a benchmark suite. Two structural issues invalidate this form as a quantitative predictor.

The scalar composite is not empirically identifiable. The four components of Eq. (1) have different scales and, in some cases, different units. In particular, the topological estimator used in Part I has units of length on the data manifold (Carlsson, 2009; Edelsbrunner and Harer, 2010). Normalizing the final sum by C_{max} rescales the output but does not repair the internal mismatch: whichever component has the largest variance over the benchmark dominates the composite, independently of whether that component drives learning difficulty.

The components are also statistically dependent. Entropy and intrinsic dimension co-vary, while correlation order and topological complexity co-vary on most natural datasets. Consequently, a linear combination of the components is not identifiable from a finite benchmark: multiple sets of weights $\{\lambda_i\}$ can produce essentially indistinguishable values of C_{data} .

Part II can test both objections directly because every component of Eq. (1) was evaluated on the 27 classical-track datasets. This analysis is exploratory rather than a pre-specified hypothesis, but it tests

whether the proposed scalar has the empirical structure required for its intended use.

The two halves of Table 1 expose the problem. With the topological component as deployed, C_{top} carries 100% of the composite variance and $\rho(C_{\text{data}}, C_{\text{top}}) = 0.998$: the composite is, empirically, almost exactly the topology component. Correcting that estimator (Appendix A) transfers the domination to entropy, which then carries 100% of the composite variance with $\rho = 0.993$. The two resulting rankings agree only at $\rho = 0.579$.

This transfer does not establish that entropy is the informative pillar. Differential entropy is extensive in dimension and scale-dependent, $h(aX) = h(X) + d \log |a|$, and therefore carries the same kind of scale defect. Across the suite,

$$S \sim d \log(\text{diam}), \quad R^2 = 0.999,$$

with Pearson $r = +1.000$. The largest value in the suite, $S = 4036$ nats for **real-fashion-2v6**, occurs for the dataset with the largest feature dimension ($d = 784$); the second largest occurs for the dataset with the largest diameter (1.2×10^6). Thus two of the four pillars inherit the same fundamental problem: the scalar sum reports whichever uncontrolled scale the current estimators amplify.

Rank normalization separates the pillars without choosing their weights. The appropriate repair is therefore to preserve the components rather than collapse them into a weighted scalar. For each component f_i and dataset D in a benchmark suite $\mathcal{B}$, we define

$$v_i(D) = \frac{|\{D' \in \mathcal{B} : f_i(D') \leq f_i(D)\}|}{|\mathcal{B}|} \in [0, 1]. \quad (2)$$

The vector

$$\mathbf{v}(D) = (v_1, \dots, v_K)$$

is the dataset’s complexity profile. Each component is now dimensionless, scale-invariant, and retained separately rather than being assigned an unidentifiable weight. The empirical CDF of each f_i is computed once on $\mathcal{B}$ and frozen; new datasets are placed in $\mathbf{v}$ -space by interpolation against those frozen CDFs.

As a diagnostic, applying Eq. (2) makes the four components carry approximately equal variance shares, 25.00%, 25.00%, 24.99%, and 25.00%, as expected from their common rank scale. This does not make the pillars equally informative; it removes the arbitrary dominance created by their native scales so that their predictive contributions can be tested separately.

Table 2 gives the resulting vectors for the 27 classical-track datasets, which are the representation consumed by the selector. The underlying estimator values are reported in Appendix B; the topology estimator is subject to the caveat in Appendix A.

A fifth pillar: geometric curvature. For data with native graph or hypergraph structure, we additionally specify a discrete Ricci-curvature summary. For graphs we use mean Ollivier–Ricci curvature (Ollivier, 2009; Ni et al., 2019),

$$\bar{\kappa}_{\text{OR}}(D) = \frac{1}{|E|} \sum_{xy \in E} \left(1 - \frac{W_1(m_x, m_y)}{d(x, y)} \right), \quad (3)$$

where m_x and m_y are probability measures on the one-step neighbourhoods, W_1 is the Wasserstein-1 distance, and d is the shortest-path distance. For hypergraphs we specify ORCHID node-transport curvature (Coupette et al., 2023), with the edge-Ricci construction (Hacquard, 2026) on the line graph when hyper-edge cardinalities span more than two orders of magnitude.

The motivation is complementary to persistent homology: topology captures global connectivity, whereas curvature captures local geometric distortion. The curvature component is also motivated by emerging quantum subroutines for its estimation, including approaches based on block-encoded geodesic distances, QSVT, and quantum power methods (Nghiem et al., 2025).

Scope note. The curvature pillar is specified but not implemented in Part II. Every dataset in the frozen suite—classes A, C, and D point clouds and class B state ensembles—is non-graph. Consequently v_5 is dropped throughout the present analysis. H2’s curvature arm is therefore inapplicable here, and H2 reduces to a test of the topological contribution alone. The unexercised pillar is recorded explicitly rather than silently omitted; it becomes testable when graph or hypergraph datasets enter the benchmark.

The intrinsic complexity and embedding interaction are different objects. Equation 11 of Part I introduced the induced quantum complexity $C_{\text{ind}}(D; \phi)$ and correctly observed that the perceived complexity of a dataset can depend on the feature map U_ϕ . The error was to treat this induced quantity as a property of the dataset itself. It is a property of the *dataset–embedding pair*.

We therefore separate the two. The vector $\mathbf{v}(D)$ contains embedding-free properties computed from the raw data. The interaction vector

$$\mathbf{q}(D, \phi) = (q_1, \dots, q_M) \in \mathbb{R}^M \quad (4)$$

contains quantities evaluated after embedding, including average and variance of bipartite entanglement entropy, the embedded-kernel effective rank $r_{\text{eff}}(K_\phi)$, circuit-family expressibility $\mathcal{E}(U_\phi)$, embedding locality/lightcone width, and the persistent-homology profile of the embedded point cloud under fidelity-induced distance.

The distinction has three consequences. First, dataset complexity can be computed once and reused across candidate embeddings. Second, feature-level

Table 1: Part I’s composite C_{data} (Eq. (1), equal weights) evaluated on the 27 classical-track datasets. The four pillars differ by up to six orders of magnitude in range and do not share units. The composite is therefore dominated by whichever pillar has the largest empirical variance, and which pillar dominates depends on the estimator. *Exploratory.*

Pillar	with C_{top} as deployed		with C_{top} corrected	
	share of Var	$\rho(C_{\text{data}}, \cdot)$	share of Var	$\rho(C_{\text{data}}, \cdot)$
S (entropy, nats)	0.00%	+0.573	100.00%	+0.993
I_{corr} (nats per order)	0.00%	−0.419	0.00%	+0.272
K (compression, dimensionless)	0.00%	−0.539	0.00%	−0.333
C_{top}	100.00%	+0.998	0.00%	−0.645
$\rho(C_{\text{data}}^{\text{deployed}}, C_{\text{data}}^{\text{corrected}}) = 0.579$				

analysis can be defined through a marginal signature $\sigma(x_j) \subset \mathbf{v}$, allowing H4 to ask how individual features interact with candidate embeddings. Third, predictor attributions can distinguish variation associated with the dataset from variation associated with its embedding.

The edge is therefore modelled as

$$\hat{e}(D, \phi) = \mathcal{S}(\mathbf{v}(D), \mathbf{q}(D, \phi)). \quad (5)$$

Two predictors are fit in parallel and both are reported: a Gaussian process with automatic relevance determination (Rasmussen and Williams, 2006), providing length-scales as feature-importance estimates and posterior variance as uncertainty; and a gradient-boosted model with quantile regression and SHAP attributions (Lundberg and Lee, 2017). Agreement between the two is evidence of robustness; disagreement is itself informative.

The cost of this revision is that no single scalar summarizes dataset complexity. The benefit is that the representation is empirically identifiable and preserves uncertainty and component-level attribution.

3.2 Correcting the barren-plateau scaling law

Part I Eq. 9 proposed

$$\text{Var}[g(\theta)] \approx \exp(-\alpha nd \tilde{C}_{\text{data}}). \quad (6)$$

The equation has an undesirable data-free limit: $\tilde{C}_{\text{data}} \rightarrow 0$ implies $\text{Var}[g(\theta)] \rightarrow 1$. This treats the absence of data complexity as removing the barren plateau altogether, whereas the Haar-random circuit result of McClean et al. (2018) establishes exponential gradient suppression as a property of the circuit ensemble itself, independent of a particular data distribution.

The corrected formulation therefore separates the circuit baseline from the data-dependent modulation:

$$\log \text{Var}[g(\theta)] \approx \log A - \alpha nd - nd \boldsymbol{\kappa}^\top \mathbf{v}(D), \quad (7)$$

where A and α describe the data-free circuit family and $\boldsymbol{\kappa} \in \mathbb{R}_{\geq 0}^K$ contains component-wise data sensitivities. At $\mathbf{v} \rightarrow \mathbf{0}$,

$$\text{Var}[g] \rightarrow Ae^{-\alpha nd},$$

recovering the circuit-only baseline. The original equation is recovered as a special case in which one complexity component dominates.

H1 tests whether Eq. (7) describes the observed scaling and whether any κ_i are non-negligible. H2 tests whether topology contributes beyond entropy, correlation, and compressibility. The empirical scaling results are reported in Sec. 6.6.

3.3 Replacing scalar expressibility–complexity alignment

Part I Eq. 15 proposed

$$\mathbb{E}[\epsilon_{\text{gen}}] \approx \epsilon_{\text{emp}} + \lambda |\mathcal{E}(U) - C_{\text{data}}|. \quad (8)$$

The construction contains two problems. Expressibility $\mathcal{E}(U)$ is a Kullback–Leibler divergence between fidelity distributions, using the Haar-random distribution as the reference (Sim et al., 2019), while C_{data} is normalized on $[0, 1]$. Their absolute difference therefore has no invariant interpretation, and the proposed “sweet spot” $\mathcal{E}(U) \approx C_{\text{data}}$ is not a well-defined testable condition.

Part II replaces the scalar alignment distance with a learned alignment score. The predictor has two heads sharing a representation $\Phi(\mathbf{v}, \mathbf{q})$:

$$\hat{e}(D, \phi) = \mathcal{S}_{\text{edge}}(\Phi(\mathbf{v}(D), \mathbf{q}(D, \phi))), \quad (9)$$

$$a(D, U) = \sigma(\mathcal{S}_{\text{align}}(\Phi(\mathbf{v}(D), \mathbf{q}(D, U)))) \in (0, 1), \quad (10)$$

where σ is the logistic sigmoid.

This formulation retains the substantive claim of Part I that mismatch between data structure and circuit representation should affect generalization without imposing an arbitrary common numerical scale. The generalization model becomes

$$\mathbb{E}[\epsilon_{\text{gen}}] \approx \epsilon_{\text{emp}} + \beta(1 - a(D, U)), \quad (11)$$

with $\beta \geq 0$ parameterized as $\beta = e^b$.

For each (D, ϕ) , the observed budget-matched edge e_{obs} and empirical generalization gap $\text{gap}_{\text{obs}} = \epsilon_{\text{test}} -$

Table 2: The intrinsic complexity vector $\mathbf{v}(D)$ after rank normalisation (Eq. (2)), for the 27 classical-track datasets — the representation the selector consumes. Each column is the empirical CDF of that component over the frozen suite, so entries are comparable across columns by construction. H differential entropy (kNN-KL); d_{pca} , $d_{2\text{nn}}$ intrinsic dimension (PCA-effective, TwoNN); K compression ratio (gzip); T topological complexity as pre-specified (Vietoris-Rips, defective — see Appendix A); T^* its corrected replacement (untruncated Rips, H_0 excluded, diameter-normalised ℓ_2); I_2 , I_h second- and higher-order mutual information (kNN, MINE); R kernel effective rank (RBF, median heuristic); M MDL (autoencoder).

Dataset	H	d_{pca}	$d_{2\text{nn}}$	K	T	T^*	I_2	I_h	R	M
adv-easy-looking-hard	0.70	0.22	0.85	0.96	0.70	0.44	0.26	0.30	0.48	0.56
adv-encoded-random	0.89	0.19	0.41	0.37	0.93	0.19	0.33	0.33	0.30	0.85
adv-hard-looking-easy	0.33	0.26	0.04	0.89	0.19	0.41	1.00	1.00	0.22	0.19
adv-quantum-trivial	0.37	0.44	0.15	0.74	0.15	0.70	0.89	0.93	0.19	0.26
real-breast-cancer	0.85	0.07	0.37	0.33	0.89	0.11	0.63	0.81	0.26	0.81
real-credit-default	0.96	0.33	0.48	0.11	1.00	0.22	0.67	0.78	0.56	0.70
real-diabetes	0.74	0.11	0.44	0.19	0.85	0.15	0.44	0.48	0.52	0.67
real-digits-3v5	0.93	0.67	0.93	0.04	0.81	0.52	0.41	0.44	0.93	0.93
real-fashion-2v6	1.00	0.59	1.00	0.07	0.96	0.37	0.59	0.63	1.00	1.00
real-higgs	0.78	1.00	0.96	0.41	0.74	0.56	0.37	0.37	0.96	0.78
real-ionosphere	0.41	0.70	0.89	0.30	0.37	0.30	0.85	0.74	0.81	0.89
real-iris	0.56	0.15	0.30	0.15	0.04	0.26	0.74	0.85	0.37	0.30
real-sonar	0.04	0.63	0.74	0.26	0.30	0.33	0.48	0.52	0.89	0.96
real-wine	0.81	0.04	0.26	0.22	0.78	0.04	0.52	0.67	0.04	0.74
synth-2sphere	0.48	0.41	0.11	0.70	0.11	0.67	0.81	0.70	0.15	0.07
synth-3sphere-pc	0.52	0.48	0.33	0.81	0.33	0.74	0.56	0.56	0.41	0.11
synth-anisotropic-gmm	0.63	0.56	0.81	1.00	0.41	0.07	0.93	0.89	0.44	0.33
synth-concentric-rings	0.30	0.30	0.07	0.78	0.07	0.48	0.96	0.96	0.07	0.22
synth-ea-spin-glass	0.26	0.78	0.59	0.52	0.52	0.93	0.19	0.22	0.63	0.41
synth-klein-bottle	0.44	0.52	0.19	0.85	0.22	0.78	0.78	0.59	0.33	0.15
synth-parity-2	0.15	0.85	0.52	0.63	0.48	0.96	0.22	0.19	0.74	0.63
synth-parity-3	0.07	0.81	0.56	0.52	0.67	0.81	0.07	0.07	0.67	0.52
synth-parity-4	0.19	0.96	0.70	0.56	0.59	1.00	0.30	0.04	0.78	0.48
synth-parity-5	0.11	0.93	0.63	0.44	0.63	0.89	0.15	0.15	0.59	0.37
synth-prng	0.67	0.74	0.78	0.67	0.44	0.63	0.11	0.26	0.85	0.59
synth-sk-spin-glass	0.22	0.89	0.67	0.59	0.56	0.85	0.04	0.11	0.70	0.44
synth-torus	0.59	0.37	0.22	0.93	0.26	0.59	0.70	0.41	0.11	0.04

ϵ_{train} provide the training targets:

$$\mathcal{L} = \sum_{D,\phi} (e_{\text{obs}} - \hat{e})^2 + \gamma \sum_{D,U} [\text{gap}_{\text{obs}} - \epsilon_{\text{emp}} - \beta(1-a)]^2, \quad (12)$$

with γ selected on a held-out fold. Thus the alignment score is not defined from itself: its training signal comes from observed generalization gaps, while on held-out datasets a is computed from $(\mathbf{v}, \mathbf{q})$ alone.

For the GP this is implemented as a multi-output GP with a linear model of coregionalization; for the GBM, as a multi-output regressor with shared splits. H3 tests $\beta > 0$ using a bootstrap confidence interval.

3.4 Making encoding cost and data access explicit

Part I described an encoding-expressivity trade-off but did not include encoding overhead in the comparison. This matters particularly for high-expressivity encodings whose state-preparation cost depends on strong data-access assumptions. Amplitude encoding, for example, can require substantial state-preparation

resources in the absence of an appropriate coherent data-loading primitive; QRAM is precisely the assumption that provides coherent address-indexed access to stored classical data in superposition (Giovannetti et al., 2008).

For an encoding ϕ on n qubits we therefore define

$$\mathcal{C}_\phi = D_\phi W_\phi (1 - \epsilon_\phi)^{-D_\phi W_\phi}, \quad (13)$$

where D_ϕ is state-preparation depth, W_ϕ is the average number of two-qubit gates per layer, and ϵ_ϕ is the reference per-gate error rate. The exponential factor represents compounding fidelity loss under the reference noise model.

We report three versions of the edge: $\hat{e}_{\text{raw}}$, the unadjusted gap; $\hat{e}_{\text{budget}}$, the gap after matching the reference resource budget; and $\hat{e}_{\text{Pareto}}$, the hypervolume of the performance-cost Pareto frontier. The primary target is $\hat{e}_{\text{budget}}$: a large raw edge with a small budget-matched edge indicates that the apparent advantage is attributable substantially to resource asymmetry.

Operationalizing the budget-matched edge.

Part II uses noiseless simulation, so the resource

penalty is introduced through the fidelity predicted by a frozen reference noise model, $\epsilon_2 = 10^{-3}$. For a cell with n_{2q} two-qubit gates,

$$F = (1 - \epsilon_2)^{n_{2q}}.$$

The noiseless quantum accuracy $a_q = 1 - \epsilon_{\text{test}}^q$ is shrunk toward chance by F , giving

$$\begin{aligned} \hat{e}_{\text{budget}} &= \left[\frac{1}{2} + F \left(a_q - \frac{1}{2} \right) \right] - a_c \\ &= \left(\epsilon_{\text{test}}^c - \frac{1}{2} \right) - F \left(\epsilon_{\text{test}}^q - \frac{1}{2} \right). \end{aligned} \quad (14)$$

This reduces to $\hat{e}_{\text{raw}}$ at $F = 1$ and approaches the classical chance-referenced term as $F \rightarrow 0$.

Data-access modes. The cost of an encoding cannot be interpreted independently of the data-access model. We therefore classify each (D, ϕ) pair as:

- **Mode S (sparse/structured):** the encoding exploits explicit structure in the data, such as sparsity or a structured sampling primitive;
- **Mode Q (QRAM-dependent):** the encoding assumes coherent random-access loading of otherwise unstructured classical data;
- **Mode N (quantum-native):** the input is a physical quantum state rather than a classically stored dataset.

This classification determines which classical comparison is admissible. For a Mode-Q pair, the classical comparator must include an appropriate quantum-inspired algorithm under the corresponding sampling/query access model. Otherwise the comparison gives the quantum procedure an access advantage that has not been charged to the quantum side. The dequantization literature makes this issue explicit: when a classical algorithm is given the corresponding sampling access, apparent quantum speedups for some low-rank linear-algebra tasks can disappear (Tang, 2019; Chia et al., 2020).

A Mode-Q result can therefore survive as a candidate quantum advantage only when no applicable classical counterpart exists under the matched access model. Such cases receive a *sparse-survivor* tag and are compared against the best unconstrained classical method only after the access assumption has been made explicit.

For Mode-Q pairs the resource model includes the active data-loading overhead,

$$\mathcal{C}_\phi^{(Q)} = (D_\phi + n)W_\phi(1 - \epsilon_\phi)^{-(D_\phi + n)W_\phi}, \quad (15)$$

whereas Eq. (13) applies to Mode S and $\mathcal{C}_\phi = 0$ to Mode N.

The present frozen suite contains no encoding requiring coherent QRAM loading. All evaluated pairs are therefore Mode S or Mode N (Sec. 6.11), and H6 is a scope check rather than an empirical test of dequantization.

4 Pre-specified hypotheses

The revisions to the framework in Sec. 3 define six pre-specified hypotheses (Tab. 3). These hypotheses were fixed before the experimental analysis, together with their decision rules, predictor architecture, encoding library, data-access classifications, and stopping criteria. The complete protocol, frozen estimator code, and analysis pipelines are released with the accompanying code. We committed in advance to reporting the outcome of every test independently of its direction.

The battery is structured hierarchically. H1–H3 test the principal mechanisms introduced by the revised framework: data-dependent trainability, geometric structure, and expressibility–data alignment. H4 tests a downstream consequence of the interaction between data characteristics and encoding choice. H5 is the central integrative test: whether these quantities, taken together, provide a predictive selector for the observed quantum edge on previously unseen datasets. H6 tests the scope of that prediction with respect to the data-access model.

Model comparison: Several hypotheses use the Akaike Information Criterion (AIC),

$$\text{AIC} = 2k - 2 \ln \hat{L},$$

where k is the number of free parameters and $\hat{L}$ is the maximized likelihood. Lower AIC indicates a preferred model. We define

$$\Delta \text{AIC} = \text{AIC}_{\text{without}} - \text{AIC}_{\text{with}},$$

so that positive values favour inclusion of the tested component. A threshold of $\Delta \text{AIC} = 10$ is used as the pre-specified criterion for substantive support (Burnham and Anderson, 2002). Comparisons are made on the same observations.

H1: Barren-plateau scaling. The revised scaling law in Eq. (7) is expected to describe the observed gradient-variance scaling across the suite. The hypothesis is supported if the cross-validated linear regression of $\log \text{Var}[g(\theta)]$ on $(n, d, nd \cdot \mathbf{v})$ achieves $R^2 \geq 0.7$ and at least one data-dependent coefficient κ_i has a 95% confidence interval excluding zero. This tests whether the revised framework captures a systematic modulation of trainability by dataset complexity beyond the data-independent circuit contribution.

H2: Topological and curvature contribution. Adding the topological and/or curvature components

Table 3: The six pre-specified hypotheses, decision rules, and outcomes. H5 is the primary integrative test of selector validity. H6 is a boundary condition on data-access mode: the frozen suite contains no Mode-Q instances, so this stratum cannot be evaluated.

	Claim	Decision rule	Outcome
H1	Barren-plateau scaling, Eq. (7)	$R^2 \geq 0.7$; some κ_i CI excludes 0	FAILS
H2	Topology/curvature contribute	$\Delta\text{AIC} \geq 10$ vs. reduced $\mathbf{v}$	FAILS
H3	Expressibility–data alignment	$\beta > 0$, 95% CI excludes 0; $\rho \geq 0.3$	Formal criterion met[†]
H4	Per-feature encoding optimality	AUROC gain $\geq 5\%$, CI excludes 0	Untested (downstream of H5)
H5	Selector validity (primary)	$R^2 \geq 0.4$ on $\hat{e}_{\text{budget}}$, ECE ≤ 0.1	FAILS
H6	Dequantization-stratified validity	conditional on H5	Not evaluable

[†] The pre-specified numerical criterion is satisfied, but the result does not provide evidence for the underlying alignment hypothesis; see Sec. 6.9.

of $\mathbf{v}$ is expected to improve the explanatory model for gradient behaviour beyond entropy, correlation, and compressibility. The hypothesis is supported if the augmented model achieves $\Delta\text{AIC} \geq 10$ relative to the reduced model. This operationalizes the geometric extension introduced in Sec. 3. The classical precedent is Andreeva et al. (2024), who study whether the topology of the optimization trajectory provides information about generalization beyond traditional statistical capacity measures. H2 asks whether an analogous topological quantity contributes to predicting *gradient* behaviour on the quantum side.

H3: Expressibility–data alignment. The fitted alignment coefficient β in Eq. (11) is expected to be positive, with a 95% confidence interval excluding zero. As a secondary criterion, the Spearman correlation between predicted $a(D, U)$ and observed $-\text{gap}_{\text{obs}}$ on held-out folds is required to satisfy $\rho \geq 0.3$. Confidence intervals are obtained by bootstrapping over leave-one-dataset-out folds.

H4: Per-feature encoding optimality. On datasets containing heterogeneous feature structure, mixed-encoding ansätze that route each feature x_j to its predicted-optimal encoding $\phi^*(x_j)$ are expected to outperform monolithic encodings. Support requires a relative AUROC improvement of at least 5% with a 95% confidence interval excluding zero. Because this test depends on the validity of the upstream selector, it is not evaluated when H5 fails.

H5: Selector validity (central test). The revised data-centric framework is expected to predict the observed quantum edge on previously unseen datasets. Support requires the trained selector $\mathcal{S}$ to achieve $R^2 \geq 0.4$ on $\hat{e}_{\text{budget}}$ under leave-one-dataset-out cross-validation, together with expected calibration error (Guo et al., 2017) of at most 0.1.

H5 is the primary test of the framework as an operational predictive model. Its failure means that the quantities specified by the revised framework, in their

present form, do not provide a reliable predictor of the measured edge across datasets.

H6: Dequantization-stratified selector validity.

The validity of the selector is additionally expected to depend on the data-access model. Conditional on H5 passing globally, performance is evaluated separately across the available access modes. The pre-specified criterion requires $R^2 \geq 0.4$ within both the Mode-S/Mode-N (including sparse-survivor) regime and the Mode-Q regime, or a significant degradation of at least $\Delta R^2 \geq 0.2$ in Mode-Q. The latter outcome would indicate that the selector distinguishes regimes in which classical access models can or cannot reproduce the relevant computation.

The frozen suite contains no Mode-Q instances, however. H6 is therefore a scope check rather than an evaluable hypothesis in the present experiment. This limitation is itself relevant to the interpretation of any apparent quantum edge, since data-access assumptions determine whether a quantum procedure has a meaningful classical counterpart.

Together, H1–H6 form the pre-specified falsification battery. The hierarchy is deliberate: H1–H4 test individual mechanisms of the revised framework, whereas H5 asks whether those mechanisms combine into a predictive model of quantum edge. H6 then places that prediction on the boundary defined by the data-access model.

This structure also separates two distinct outcomes. A component hypothesis may fail because the proposed mechanism is absent or misspecified, while H5 asks the stronger question of whether the complete framework succeeds as a predictive model. The primary conclusion of Part II therefore rests on H5, while the preceding tests identify which mechanisms prevent the proposed data-centric predictor from becoming a reliable measure of quantum advantage.

5 Methodology

We evaluate the proposed data-centric predictor under a fully pre-specified experimental protocol. The confirmatory analysis uses a frozen benchmark registry, a fixed encoding-ansatz grid, repeated random seeds, and leave-one-dataset-out (LODO) validation. All constructions, hyperparameters, circuit families, preprocessing rules, and decision criteria were fixed before evaluating the held-out predictive performance.

5.1 Benchmark suite

The frozen registry contains four dataset classes, of which Classes A, C, and D comprise the 27-dataset confirmatory Phase B suite analyzed in this paper. These datasets span heterogeneous binary classification problems while remaining within a common classical-data access model.

For each dataset, the experimental protocol evaluates ten deterministic quantum encodings, six variational ansatz configurations, and five random seeds. The confirmatory grid therefore contains

$$27 \times 10 \times 6 \times 5 = 8100 \quad (16)$$

dataset-encoding-ansatz-seed cells. All 8100 cells were completed.

The quantum circuits operate on eight qubits. The ten encodings comprise four encoding families: angle encoding, ZZ-type encoding, amplitude encoding, and data re-uploading. The six ansatz configurations comprise three circuit families evaluated at depths $L \in \{1, 3\}$. The same experimental grid and completed cells are used throughout the confirmatory analysis; LODO validation partitions these observations by dataset rather than introducing a new experimental sample.

5.2 Experimental design

The encoding procedure is deterministic: for a given input dataset and encoding configuration, the same data point is mapped to the same quantum representation. The encoding itself is therefore not randomized.

The five seeds randomize four components of the experimental pipeline: the train-test split, the random state of the logistic-regression solver, the subsampling used to estimate the interaction vector $\mathbf{q}(D, \phi)$, and the parameter initialization of the quantum classifier. Because the train-test split changes with the seed, the set of points presented to the encoding also changes. Encoding results are therefore cached with the seed as part of their key, despite the encoding map itself being deterministic.

For every dataset-encoding-ansatz-seed cell, we record the resulting quantum and classical performance together with the quantities required by the

proposed predictor. These measurements form the common input table for both the Gaussian-process and gradient-boosting analyses.

5.3 Measured quantities and prediction target

For each cell (D, ϕ, s) , we record the quantum accuracy a_q , the classical reference accuracy a_c , empirical and generalization errors, gradient statistics, encoding cost, the normalized data representation $\mathbf{v}(D)$, and the circuit-data interaction vector $\mathbf{q}(D, \phi)$.

The primary prediction target is the budget-matched quantum edge. We model a fixed two-qubit error rate

$$\epsilon_2 = 10^{-3} \quad (17)$$

and define the corresponding survival factor for a circuit containing n_{2q} two-qubit gates as

$$F = (1 - \epsilon_2)^{n_{2q}}. \quad (18)$$

The budget-matched quantum accuracy is then

$$a_q^{\text{budget}} = \frac{1}{2} + F \left(a_q - \frac{1}{2} \right), \quad (19)$$

and the primary edge is

$$e_{\text{budget}} = a_q^{\text{budget}} - a_c. \quad (20)$$

Thus, resource matching modifies the quantum performance while leaving the classical comparator unchanged.

The raw edge,

$$e_{\text{raw}} = a_q - a_c, \quad (21)$$

and the Pareto-based edge are retained as secondary analyses, but the budget-matched edge in Eq. (20) is the target used for the primary integrative prediction test.

The predictor has two output heads. The first predicts the edge,

$$\hat{e} = \hat{e}(D, \phi), \quad (22)$$

while the second stores the predicted generalization gap directly. The observed gap is defined as

$$g_{\text{obs}} = \epsilon_{\text{gen}} - \epsilon_{\text{emp}}, \quad (23)$$

and the predictor returns a corresponding estimate $\hat{g}$. No empirical error is reintroduced as an independent predictor target.

5.4 Predictor representation and architecture

The data representation is constructed from rank-normalized data descriptors and circuit-data interaction quantities. For each descriptor f_i , its value is converted to an empirical rank within the training portion of the benchmark:

$$v_i(D) = \frac{|\{D' : f_i(D') \leq f_i(D)\}|}{|\mathcal{B}|}, \quad v_i(D) \in [0, 1]. \quad (24)$$

The resulting vector $\mathbf{v}(D)$ is combined with the interaction vector $\mathbf{q}(D, \phi)$ to form the joint representation

$$\mathbf{z}(D, \phi) = [\mathbf{v}(D), \mathbf{q}(D, \phi)]. \quad (25)$$

The interaction vector contains measurements of circuit properties relevant to the proposed mechanism, including entanglement, kernel effective rank, expressibility, locality/lightcone width, and persistent-homology descriptors of the embedded point cloud.

The confirmatory predictor is implemented as a sparse variational Gaussian process with 256 inducing points and a linear model of coregionalization (LMC) variational strategy over the two prediction heads. An exact Gaussian process is used only as a frozen reference implementation because an exact fit is computationally prohibitive at the full sample size. On an identical set of 800 training inputs, the sparse implementation reproduces the exact-GP predictions with Pearson correlation $r = 0.984$ and RMSE = 0.026. This comparison is an implementation-fidelity check rather than an independent measure of predictive generalization.

A gradient-boosting model is trained in parallel using the same representation and the same outer validation partitions. This provides a non-Gaussian reference model while holding the predictor inputs and cross-validation structure fixed.

5.5 Cross-dataset validation

Predictive performance is evaluated using leave-one-dataset-out cross-validation over the 27 datasets. In each of the 27 folds, all cells belonging to one dataset are held out, while the remaining 26 datasets provide the training set. The Gaussian-process and gradient-boosting predictors use identical outer train-test partitions.

All transformations that estimate dataset-level ranks are fitted using only the outer training rows and then applied to the held-out rows. The raw $\mathbf{v}(D)$ values are therefore generated before the fold-level transformation, but no information from the held-out dataset enters the rank normalization used by the predictor. Hyperparameters that require model selection, including the loss weighting parameter γ , are selected within the training portion of each outer fold using inner LODO validation.

The predictor is evaluated only after the complete outer training procedure has been fitted. Predictive performance is assessed on the held-out dataset in each LODO fold using the coefficient of determination and expected calibration error (ECE). For the primary edge prediction, the coefficient of determination is computed as

$$R^2 = 1 - \frac{\sum_{j \in \mathcal{T}_{\text{test}}} (e_j - \hat{e}_j)^2}{\sum_{j \in \mathcal{T}_{\text{test}}} (e_j - \bar{e}_{\text{test}})^2}, \quad (26)$$

where e_j is the observed budget-matched edge, $\hat{e}_j$ is the corresponding prediction, and $\bar{e}_{\text{test}}$ is the mean observed edge over the held-out test set.

The ECE measures calibration of the predictor’s uncertainty estimates by comparing predicted and empirical coverage across the held-out observations. The pre-specified H5 acceptance criterion requires

$$R^2 \geq 0.4 \quad \text{and} \quad \text{ECE} \leq 0.1. \quad (27)$$

This procedure tests cross-dataset generalization rather than interpolation across cells from the same dataset. Because the 8100 experimental cells are fixed before the LODO procedure, each fold partitions the same completed experimental dataset into training and held-out datasets.

5.6 Classical baselines and attribution analysis

The primary classical comparator is logistic regression trained on the original classical features using the same train-test partition as the quantum model. Its accuracy defines the primary classical reference,

$$a_c = a_{\text{LR}}, \quad (28)$$

and therefore determines both the raw and budget-matched primary edges.

As a sensitivity analysis, we evaluate a fixed library of six classical classifiers:

1. logistic regression, with $C = 1.0$ and $\text{max_iter} = 2000$;
2. k -nearest neighbors, with $k = 5$;
3. support-vector classification with an RBF kernel, $C = 1.0$, and $\gamma = \text{scale}$;
4. a multilayer perceptron with hidden layers $(64, 64)$, $\text{max_iter} = 800$, and $\alpha = 10^{-3}$;
5. histogram gradient boosting with default hyperparameters;
6. random forest with 300 trees and a single worker.

All six models operate directly on the original classical features, use the same training and test partitions as the quantum experiment, and retain their fixed hyperparameters throughout the analysis.

Let $\mathcal{M}_6$ denote this library. For each dataset and seed, we define the best-of-six classical accuracy as

$$a_{6\text{CL}} = \max_{m \in \mathcal{M}_6} a_m. \quad (29)$$

The corresponding sensitivity edges are

$$e_{6\text{CL}, \text{raw}} = a_q - a_{6\text{CL}} \quad (30)$$

and

$$e_{6\text{CL}, \text{budget}} = a_q^{\text{budget}} - a_{6\text{CL}}. \quad (31)$$

These quantities are not used to redefine the primary H5 target. Instead, the paired comparison tests whether predictor signal persists when the classical comparator has substantially greater model capacity. A signal that disappears against the best-of-six comparator is consistent with the predictor tracking failure of the logistic-regression baseline rather than a robust quantum advantage.

For binary classification, chance accuracy is

$$a_{\text{chance}} = \frac{1}{2}. \quad (32)$$

Chance is used as a reference for determining whether a model learns a predictive signal; it is not a competing classical baseline and is not subtracted from the primary quantum edge.

Classical baselines are trained on the original, unencoded features. This is the primary comparison: the classical comparator receives the raw data, whereas the quantum model receives the data after the specified encoding. Where we analyze the effect of the encoding itself, we additionally compare models under the same encoded representation; these are attribution analyses rather than replacements for the primary baseline.

Finally, the framework specifies an access-model-matched comparison for settings in which the classical and quantum methods do not receive the same effective information. Under sparse or structured access (Mode S), the classical comparator may use the corresponding sparse representation. Under quantum-access assumptions (Mode Q), a valid comparison requires a classical or quantum-inspired method with the corresponding sampling or query access. This condition prevents an apparent quantum edge from arising solely from asymmetric data-access assumptions. The confirmatory suite analyzed here does not contain Mode Q cells; consequently, this comparison is treated as a framework boundary condition rather than as an empirical test in the present study.

5.7 Reproducibility and analysis freeze

All circuits, encoding maps, ansatz families, hyperparameters, random seeds, preprocessing operations, predictor architecture, and decision rules were fixed before the held-out predictor performance was evaluated. No component of the predictor or benchmark was modified in response to the held-out results.

Three distinct sources of variability are considered separately. First, seed variation captures the complete experimental pipeline, including data splitting, parameter initialization, and interaction-vector estimation. Second, the exact-versus-sparse Gaussian-process comparison quantifies implementation fidelity on identical inputs. Third, LODO cross-validation measures generalization across datasets. These three layers are not treated as interchangeable estimates of uncertainty.

Table 4: Confirmatory H5 evaluation. Pipeline: logistic-regression $\hat{e}_{\text{budget}}$, SVGP + GBM multi-head, leave-one-dataset-out over 27 folds, inner γ -CV, and three-step EM. The pre-specified threshold is $R^2 \geq 0.4$.

Metric (GP head)	Value
Pooled LODO R^2	−0.32
(corrected-gap re-run)	−0.37
Per-fold R^2 , median	−1.12
Per-fold R^2 , mean	−28.54
Fold-bootstrap 95% CI	(−58.5, −4.7)
GBM head, pooled LODO R^2	−0.58
Expected calibration error	0.60
H5 pass ($R^2 \geq 0.4$)	False

The resulting analysis therefore distinguishes variability within experimental cells, fidelity of the sparse predictor approximation, and cross-dataset predictive generalization. This separation is maintained throughout the confirmatory analysis.

6 Results

6.1 H5: the selector does not predict the edge

The selector does not predict the budget-matched edge on held-out datasets. Every aggregation falls far below the pre-specified threshold (Table 4); the fold-bootstrap upper bound of −4.7 is nowhere near 0.4. The best single fold is +0.077 (**synth-parity-3**); no fold reaches +0.1. The GBM head, an independent model on the same representation, agrees (−0.58).

We take the *pooled* R^2 as the headline metric. Per-fold R^2 is unstable here because several held-out datasets have near-zero edge variance, on which R^2 becomes extremely negative (e.g. **synth-klein-bottle** −271, **synth-3sphere-pc** −246, **synth-prng** −150). These values inflate the per-fold mean to −28.5 without carrying additional information. The pooled statistic, obtained by predicting every held-out point and scoring once, still shows that the predictor performs worse than the intercept-only mean predictor (Fig. 1).

A negative R^2 is not a weak positive result. It means that, on this suite and for this edge definition, the specified predictor does not provide a reliable cross-dataset prediction of the measured quantum edge. Under the pre-specified criterion, H5 therefore fails. The remaining results decompose this failure into the dependence of the measured edge on the classical comparator, the loss of headroom introduced before the quantum model, and the near-chance performance of the quantum model itself.

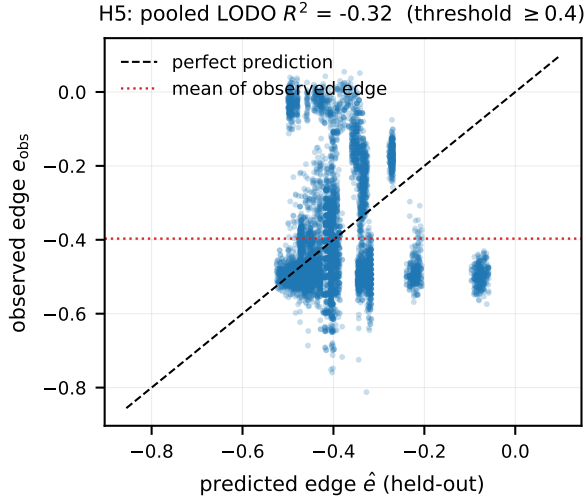


Figure 1: H5, the central test. Predicted against observed budget-matched edge on held-out datasets, pooled over all 27 leave-one-dataset-out folds. A predictive selector would concentrate along the diagonal; the cloud instead sits largely around the mean of the observed edge, corresponding to a pooled $R^2 = -0.32$.

6.2 The classical baseline determines the measured edge

The primary H5 analysis uses logistic regression as the pre-specified classical comparator. We next vary that comparator while leaving the quantum pipeline unchanged. This sensitivity analysis isolates how much of the measured edge and its predictability depends on the capacity of the classical reference.

The primary classical baselines therefore use the raw feature representation; encoded-feature comparisons are used only to isolate the contribution of the encoding.

Against logistic regression, the complexity selector gives an apparent positive held-out relationship: leave-one-dataset-out $R^2 = +0.13$, with per-dataset Spearman $\rho = 0.43$ (Fig. 2). Replacing the comparator with the best performer among six fixed classical models removes both the apparent advantage and its predictability. The six-model library consists of logistic regression, k -nearest neighbors, SVC with an RBF kernel, a multilayer perceptron, histogram gradient boosting, and random forest, with the fixed hyperparameters specified in Sec. 5.6.

All six models are trained on the same train-test partitions as the quantum experiment and are evaluated on the original classical features. They are fixed rather than tuned. The library is therefore a deliberately modest sensitivity comparator, not an estimate of the strongest classical performance achievable with unrestricted model selection.

On the full suite, the quantum win rate falls from 17% against logistic regression to 1% against the best of six. On the parity subset, where the framework

Table 5: Quantum win rate against each classical baseline taken *individually*, over the same 8,100 Phase B cells. A win is a strictly lower held-out error at matched (dataset, seed). Logistic regression is not merely the weakest of the six — it is a factor of three below the next weakest, and every other single model already places the win rate under 6%. The headline therefore does not require taking the best of the set to disappear; any of the other five would have removed it.

baseline	full suite	parity family
logistic regression	17.0%	58.3%
k -NN	5.8%	0.0%
MLP	4.8%	0.0%
SVC-RBF	4.0%	0.0%
random forest	3.6%	1.8%
HistGBM	3.5%	6.8%
worst of six	20.9%	62.4%
best of six	1.1%	0.0%

predicts an edge, the win rate falls from 59% to 0%. Correspondingly, the selector’s R^2 falls from $+0.13$ to -0.94 and per-dataset Spearman ρ from 0.43 to 0.07.

The endpoints alone could suggest that only taking the best of six removes the apparent advantage. The individual-baseline sweep shows otherwise. Table 5 reports the win rate against each of the six models on the same 8,100 cells.

Logistic regression is a clear outlier: every other individual baseline already reduces the full-suite win rate below 6%. On the parity subset, k -NN, MLP, and SVC-RBF each give a win rate of exactly 0% on their own. The apparent 58–59% quantum win rate therefore requires the logistic baseline specifically.

The mechanism is especially transparent for parity. **Parity is a canonical example of a task that a linear model cannot represent directly, while nearest-neighbour and kernel methods can capture it. A quantum win rate measured only against logistic regression can therefore reproduce the linear model’s shortfall rather than isolate a quantum contribution.** The spread across individual baselines is itself a useful diagnostic: **a large gap between one baseline and the rest should be treated as a warning before any aggregate advantage is interpreted.**

A quantum advantage reported against a single weak classical baseline does not isolate a quantum contribution. A baseline sweep is therefore required to establish whether the measured edge is robust to reasonable variation in classical model capacity.

6.3 Classical preprocessing removes headroom

The baseline analysis identifies one source of the H5 failure. A second mechanism occurs earlier in the pipeline: the classical preprocessing required to place high-dimensional data on a few-qubit register can remove the nonlinear structure that a downstream model could exploit.

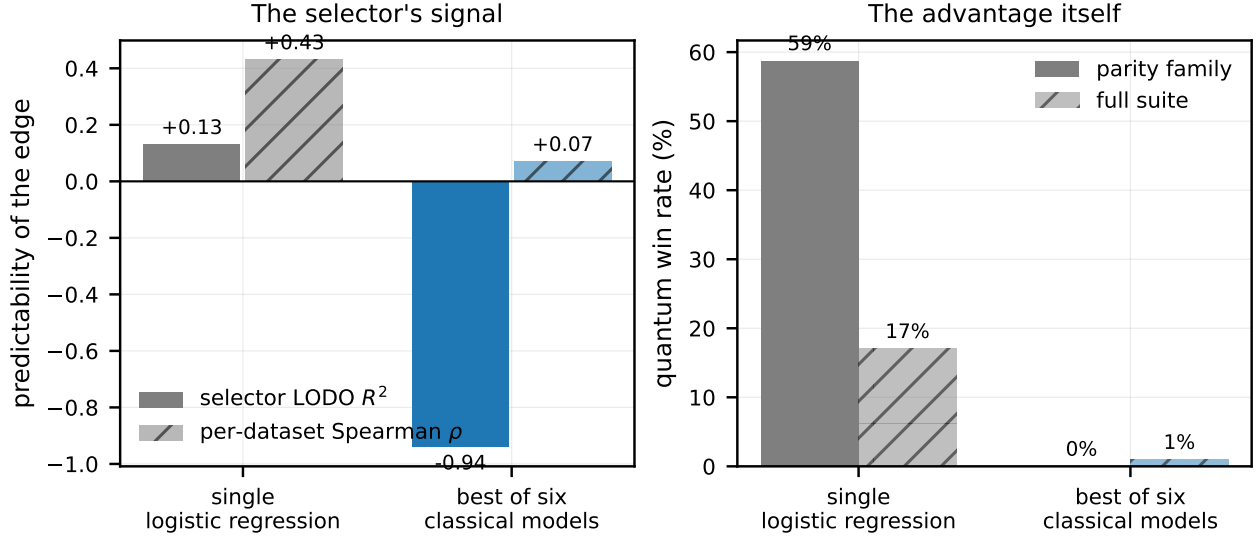


Figure 2: The classical baseline determines the measured edge. Left: the selector’s leave-one-dataset-out R^2 and per-dataset Spearman ρ on the edge. Right: the quantum win rate on the parity family (solid) and across the full suite (hatched). The quantum pipeline is unchanged; only the classical comparator changes.

Table 6: **The cost of dimensionality reduction to run on a superposition.** Upper block: garbage of the reduction map, enumerated exactly over the whole quantized input cube at matched (p, d, b) , so both reducers discard the same $p - d$ dimensions. Feature selection attains the floor $(p - d)b$ exactly in every row — it drops whole registers and touches nothing else — while an orthonormal rotation must round, and rounding destroys more than it was asked to. The gap is small in the Landauer average and $2.1\text{--}2.3\times$ in the worst-case preimage, which is what an ancilla count pays for. Lower block: reversible circuit resources at $b = 8$. Selection is a relabelling of registers; a rotation accumulates p constant-multiplies per output, needing $2b + \lceil \log_2 p \rceil$ bits of accumulator.

(p, d, b)	floor	selection	rotation	max preim.
(5, 2, 3)	9	9.000	9.220	1182
(6, 2, 3)	12	12.000	12.216	9709
(6, 3, 3)	9	9.000	9.352	2047
(7, 2, 3)	15	15.000	15.177	68475

dataset ($d=4$)	sel.	rot. anc.	garbage	overhead
breast cancer	0	84	52	$2.6\times$
wine	0	80	48	$2.5\times$
digits	0	88	56	$2.8\times$

At matched (p, d, b) , both reducers discard the same $p - d$ dimensions, so the minimum information loss associated with the discarded coordinates is fixed. Feature selection attains this floor, $(p - d)b$ bits, in every row of Table 6. An orthonormal rotation must additionally quantize the transformed coordinates, producing greater loss in the measured preimage statistics. The circuit cost is also substantially different: selection requires no gates or ancillas, whereas a rotation requires additional accumulator resources. On a 32-qubit data register, the PCA implementation requires 84 ancillas before the quantum encoder is applied.

Table 7: **Headroom after reduction, and the controls that make the zeros readable.** *Headroom* is gradient boosting minus a linear probe on the same reduced features, the accuracy a non-linear model — quantum or otherwise — could still be demonstrating something on. The maximum is taken over widths d from 2 to the full feature count, so these are not small- d artefacts. On the benchmark datasets it is indistinguishable from zero at every width, including no reduction at all. The two positive controls are what license reading that as a property of the data: the certified $[16, 4, 5]$ code data, whose label is a $\text{GF}(2)$ combination no linear-over- $\mathbb{R}$ model can see, and parity-8. **Parity-8 also shows the metric’s limit** — its headroom reads 0.004 because gradient boosting fails on parity too, so headroom near zero means *either* no exploitable structure *or* structure beyond the non-linear baseline as well.

dataset	kind	p	max headroom	d^*
breast cancer	benchmark	30	-0.004	—
wine	benchmark	13	-0.011	—
digits	benchmark	64	+0.077	4
certified $[16, 4, 5]$	control	16	+0.127	2
parity-8	control	16	+0.126	3

The reduction leaves little nonlinear headroom. Table 7 reports headroom as the performance of gradient boosting minus that of a linear probe on the same reduced features, maximized over widths from $d = 2$ to the full feature count. On breast cancer, wine, and digits, the measured headroom is indistinguishable from zero at every width, including with no reduction. PCA improves accuracy relative to selection (0.872 versus 0.812 averaged over $d \in \{2, 4, 8\}$), but this gain is already available to a linear classical model.

This provides a mechanism for the H5 null. If the classical reduction has already exposed the task to a low-capacity model, there may be little nonlinear structure left for the quantum model to exploit. This is consistent with the baseline result: the apparent edge against logistic regression is reduced when the comparator has access to nonlinear classical capacity.

Now, the classical preprocessing versus the quantum feature map. The attribution should be restricted to the classical preprocessing part of the pipeline. A classical reduction, normalization, or coordinate transformation can be handed to the classical comparator and therefore cannot itself constitute evidence of quantum advantage. This does not rule out an advantage arising from the subsequent quantum feature map: an entangling quantum map may induce a classically difficult kernel. The experiment here instead finds that, for the tested data and encoding library, the quantum prediction axis remains near chance.

The distinction can be made explicit by adding and subtracting the performance of a classical learner operating on the same preprocessed representation. The first term is obtainable by a classical party at the cost of applying the same classical preprocessing. Only the residual can support a claim specific to the quantum operation. This decomposition motivates the baseline-sensitivity analysis (see Proposition 1).

Data processing does not create label information. For data (X, Y) and an encoding channel $\mathcal{E}$ acting on X , $I(\mathcal{E}(X); Y) \leq I(X; Y)$. Thus an encoding cannot create label information absent from the input (see Proposition 2). A quantum feature map can nevertheless alter the computational difficulty of accessing information that is already present. **The relevant condition for advantage is therefore not quantum structure alone, but quantum structure aligned with label-relevant information.**

The encoder can only raise the classical floor.

The two measurements above, the information floor and the ancilla cost, describe what reduction *does*. There is a shorter statement of why it cannot do otherwise, and it is structural rather than empirical.

An encoder acting on classical data is itself a classical algorithm. Whatever it computes, a rotation, a projection, a coordinate selection, a normalization, is

computation performed on a classical machine, and therefore computation the classical comparator could also have performed. It follows that the encoder cannot lower the quantum ceiling. It can only raise the classical floor.

This is why the effect presents as structure being revealed when it is not. A good encoding makes the data separable, and separability is exactly the property that makes a task classically easy. Linear discriminant analysis is the limiting case and the reason we exclude it below: it projects onto at most $C - 1$ directions chosen to maximize between-class scatter, so whatever separation survives is close to trivial and any downstream classifier finds it. The encoder has done the work the model is then credited with.

Two things follow that we adopt as method. First, a fairness rule: **the classical baseline must be evaluated on the encoded representation, not on the raw data**. Comparing a quantum model on encoded features against a classical model on raw features charges the classical side for work the quantum side was given. Second, a scope condition: the argument holds when the encoding is a classical transformation of classical data, which is the entire setting of this paper. It does not hold when state preparation itself performs something classically hard, or when the state does not originate from classical data, a device’s output, an experimental sample, a sensor record. There the classical comparator has nothing to be handed, and the argument has no purchase.

One distinction has to be kept or the argument overreaches. It applies to the *classical preprocessing* half of an encoding pipeline, the reducers, normalizations and coordinate choices this paper measures, because those are classical maps and can be handed over intact. It does not apply to the quantum feature map itself: making the map more entangling can render the induced kernel harder to evaluate classically, and that is not a classical computation the comparator could have performed. Advantage from a feature map is not ruled out here; advantage from *preprocessing* is.

With that scope, the consequence for the program is stronger than the null results that prompted it, and it is not that these pipelines were badly designed. **The classical-preprocessing stage of a quantum pipeline cannot place an advantage beyond classical reach**, because it is classical and can always be handed over; improving it improves the baseline it is measured against. Whatever survives a comparison run on the encoded representation is attributable to the quantum map, and whatever does not was never the model’s to claim.

The complementary axis is not untested here, and its result is why this paper cannot speak to feature-map advantage either way. The Phase B sweep varies the encoding over ten configurations in four families, angle, ZZ, amplitude, re-uploading, including

genuinely entangling ZZ maps reaching 1.52 nats of half-cut entropy. The axis *collapsed*: mean a_q spans 0.462 to 0.497 across all ten, a range smaller than the within-encoding spread of 0.05–0.09, and the best encoding is scattered over nine of the ten configurations as expected from noise. Nothing distinguishes a product-state angle encoding from an entangling ZZ map because both sit at chance. So the residual term of Eq. (33) was measured and found indistinguishable from zero *in this regime*; that is a statement about the regime, not about feature maps.

Proposition 1 (attribution decomposition).

Let P be a classical preprocessing map computable in time T_P , E an encoding, Q a quantum learner and C a classical learner class, and write $\mathcal{A}(\cdot)$ for performance under a fixed metric and split, abbreviating $QEP = Q \circ E \circ P$. Then

$$\begin{aligned} & \underbrace{\mathcal{A}(QEP) - \mathcal{A}(C_{\text{raw}})}_{\text{reported gap}} \\ &= \underbrace{\mathcal{A}(CP) - \mathcal{A}(C_{\text{raw}})}_{\text{preprocessing-attributable}} + \underbrace{\mathcal{A}(QEP) - \mathcal{A}(CP)}_{\text{residual}} \end{aligned} \quad (33)$$

The identity is trivial, it adds and subtracts $\mathcal{A}(C \circ P)$, and we state it as a proposition because its content is not the algebra but the attribution it forces. The first term is obtainable by any classical party at cost T_P per example, since P is a classical map; it is therefore not available as evidence of quantum advantage. Only the residual is.

Corollary. A comparison reporting only the left-hand side does not distinguish the two terms, and so upper-bounds rather than measures the quantum-attributable part. The decomposition is exhaustive, so the question “which term did you measure?” has an answer for every published gap, and a study that cannot answer it has not isolated a quantum effect.

We adopt Eq. (33) as the reporting standard for the remainder of this series. It is also the discipline behind Sec. 6.2: the best-of-six classical baseline evaluated on the same reduced representation is precisely $\mathcal{A}(C \circ P)$, and the collapse of the measured edge when it replaces a raw-data logistic regression is the first term being paid back.

Proposition 2 (an encoding cannot manufacture label information). Let (X, Y) be the data and label random variables and let $\mathcal{E}$ be any encoding channel acting on X alone, classical preprocessing, quantum feature map, or their composition. Then

$$I(\mathcal{E}(X); Y) \leq I(X; Y), \quad (34)$$

by the data-processing inequality, which holds in the quantum case as well because $\mathcal{E}$ is a channel on

X and Y is untouched. Equality requires $\mathcal{E}$ to be sufficient for Y .

The consequence separates two things that are routinely run together. A quantum feature map *can* manufacture computational hardness: an IQP-type map induces a kernel believed hard to evaluate classically, and that hardness is a property of the map, obtainable from any input. What Eq. (34) forbids is manufacturing *relevance*. The map cannot put label information into the representation that the data did not already carry; it can only make some of what is there harder for a classical party to reach, while, by the reducer measurements above, destroying some of the rest.

An advantage therefore needs both conditions at once: the induced kernel must be classically hard *and* aligned with the structure the labels actually have. Hardness alone buys nothing, because a kernel that is expensive to compute and unrelated to the task is an expensive random kernel.

This is what the collapse of the encoding axis measures. The ZZ maps in the sweep produce genuinely entangled states, 1.52 nats of half-cut entropy against a possible 2.77, so quantum structure is present in the *states*. It is absent from the *predictions*, which sit at chance alongside the product-state angle encodings. Structure in the representation is not structure aligned with the task, and only the second is an advantage.

Why LDA is not among the reducers. Linear discriminant analysis is the standard supervised answer and it is excluded on principle rather than on cost. It projects onto $C - 1$ dimensions chosen to make the classes linearly separable, driving the headroom to zero *by construction*: a threshold suffices afterwards, and a quantum model placed on top of it is demonstrating nothing. The measurement above says PCA does the same thing on these datasets, merely less deliberately.

The controls, and the metric’s limit. A headroom instrument that reads zero on everything cannot be told from a broken one, so Table 7 carries two positive controls: the certified [16, 4, 5] code data of the companion decoding-window work, whose label is a GF(2) combination invisible to any model linear over $\mathbb{R}$, and parity-8. Both read large. Parity-8 also exposes the metric’s limit and we state it rather than bury it: its headroom is 0.004, indistinguishable from breast cancer’s, because gradient boosting fails on parity as well. **Headroom near zero therefore means either no exploitable structure or structure beyond the non-linear baseline too**, and only a known Bayes floor separates the two, which is available for generated data and not for these benchmarks.

Table 8: H1 and H2. Fit of Eq. (7) with $\mathbf{v}$ rank-normalised on training folds only; R^2 is pooled leave-one-dataset-out; coefficient intervals are cluster bootstrap over datasets (2,000 resamples). Reported coefficients are those of the design matrix, so the fitted value on nd is $-\alpha$ and on ndv_i is $-\kappa_i$.

Quantity	Value
Pooled LODO R^2 , full $\mathbf{v}$	+0.026
Pre-specified threshold	0.70
$\hat{\alpha}$ (from nd coefficient)	-0.075
κ with 95% CI excluding 0	2 of 9
higher-order MI	$\kappa = +0.052$
MDL (autoencoder)	$\kappa = +0.053$
ΔAIC , <i>topology</i> in vs. out	+0.1
Pre-specified threshold	10
LODO R^2 without topology	+0.027
H1 pass	False
H2 pass	Inconclusive [†]

6.4 H1 and H2: trainability and topology

We estimated $\text{Var}[g(\theta)]$ using the two-point parameter-shift rule on the Z_0 observable, with 200 draws of encoded state, $\theta \sim \text{U}[0, 2\pi]^P$, and parameter index per cell. The confirmatory fixed-width grid contains $27 \times 8 \times 6 \times 5 = 6,480$ cells. The complexity vectors are the Phase B vectors themselves, so the regressors are identical to those used by the selector. The width sweep reported below is exploratory and is kept separate from the confirmatory H1/H2 analysis (Table 8).

H1 fails. Equation (7) explains essentially none of the held-out variance in $\log \text{Var}[g]$: pooled LODO $R^2 = +0.026$ against a threshold of 0.70. Two of nine components, higher-order mutual information and MDL compression length, have κ intervals excluding zero and both have the predicted positive sign. The data-modulation terms are therefore detectable for these components, but their magnitude is insufficient to make the pre-specified scaling relation predictive at fixed circuit width.

The fitted relation is phenomenological rather than an exact consequence of the circuit family. **The result therefore does not establish that data complexity is absent from gradient statistics; it establishes that the specified multiplicative correction does not provide the required cross-dataset prediction on this grid.** Exact variance expressions can depend on the representation channels of the encoded state and observable, suggesting that a single scalar modulation of an nd decay rate may be too restrictive for this setting. This interpretation is post hoc and does not alter the pre-specified H1 verdict.

H2 is inconclusive. The production estimator changes AIC by only +0.1 and leaves held-out R^2 essentially unchanged. Audit of the topological instrument found that its value was dominated by

dimension-0 persistence, tracked ambient scale, used a truncated filtration, and returned zero for a perfect circle (Appendix A). The pre-specified test therefore did not measure the intended topological quantity reliably.

We rebuilt the estimator using untruncated Rips persistence, excluding H_0 and normalizing bars by point-cloud diameter, and validated it on shapes with known Betti numbers. Re-running H2 with these corrected summaries gives an estimator-dependent verdict: the threshold-free summaries fail, whereas a bar-count summary passes only in the narrow significance-cut range $\theta \in [0.08, 0.10]$ (Table 9). Because that choice is informed by the observed outcome, it cannot be promoted to a confirmatory result.

We therefore retain H2 as inconclusive and treat the repaired analysis as exploratory. Repairing an implementation defect is distinct from selecting among repaired definitions after observing the outcome. The defensible conclusion is that this experiment does not determine whether topology contributes to barren-plateau prediction. Since the frozen $\mathbf{v}$ contains no explicit curvature component, H2 reduces on this suite to a test of topology alone.

6.5 Why the primary test had little quantum variation to detect

The H5 target is informative only to the extent that the quantum model produces variation in performance beyond the classical comparator. Across the 8,100 cells, quantum test accuracy is 0.481 ± 0.069 , with median 0.493. The model is at or near chance ($a_q \geq 0.45$) in 81.5% of cells and within one percentage point of chance ($a_q \geq 0.49$) in 55.2%. In contrast, the best of the six classical models reaches $a_c \leq 0.05$ error in 68.9% of cells.

With a_q concentrated near chance, the measured edge is largely determined by the classical term:

$$e \approx \frac{1}{2} - a_c. \quad (35)$$

Consistently, the edge correlates strongly with classical accuracy, $\rho(e, a_c) = +0.913$, whereas $\rho(e, a_q) = -0.337$. Approximately 82% of the variance in the measured edge is attributable to the classical term under this decomposition. **The result explains why a data-complexity vector can appear predictive against a weak linear baseline while failing against stronger classical models.**

The null is nevertheless narrow. None of the tested alternatives predicts the edge robustly: the nine complexity components give $R^2 = -0.111$, sample size -0.138 , dimension -0.192 , their combination -0.266 , and best-classical error -0.078 . The best individual complexity component reaches only -0.003 and the best pair -0.016 . Predicting a_q itself also fails ($R^2 = -0.746$ for the linear model and -0.078 for boosting), with only $\text{sd}(a_q) = 0.020$ across datasets.

Table 9: H2 re-run on the corrected topological component ($n = 8$ grid). The production estimator reproduces the original values, while corrected summaries give an estimator-dependent result.

Topology summary	ΔAIC	κ_{top} (95% CI)	H2
production (unnormalised, H_0 in)	+0.1	+0.014 (−0.022, +0.046)	fail
corrected, ℓ_2	+0.5	+0.018 (−0.010, +0.070)	fail
corrected, max bar	−1.7	−0.004 (−0.027, +0.026)	fail
corrected, count $\theta = 0.08$	+11.5	+0.056 (+0.025, +0.093)	pass
corrected, count $\theta = 0.10$	+15.4	+0.073 (+0.042, +0.115)	pass
corrected, count $\theta = 0.15$	−1.7	−0.016 (−0.142, +0.054)	fail
<i>Threshold for a pass: $\Delta\text{AIC} \geq 10$.</i>			

Table 10: Functional form of the width dependence on the exploratory n -scaling arm. Lower AIC is better.

Model for $\log \text{Var}[g]$	R^2	AIC
$1 + nd$	+0.049	19837
$1 + n + d$	+0.164	18175
$1 + n$	+0.164	18174
$1 + \log n$	+0.171	18054
$1 + n + n^2$	+0.171	18068

The appropriate conclusion is therefore conditional rather than universal: on classical datasets for which the tested quantum model remains at chance, there is no robust quantum edge for the complexity selector to rank. **This does not establish that data complexity can never predict a quantum advantage; it establishes that the present quantities do not do so in a regime where the quantum performance itself has little variation.**

6.6 Exploratory circuit-width scaling

Because the fixed-width Phase B grid cannot identify the width dependence of Eq. (7), we collected an exploratory width-scaling arm at $n \in \{4, 6, 8, 10, 12, 14\}$ on a stratified nine-dataset subset, all five seeds, for 12,960 cells. This arm tests the functional form of the scaling relation rather than a pre-specified hypothesis.

Replacing nd by $\log n$ improves AIC by 1783, strongly favouring the log-width form within this exploratory arm. Adding depth to the width-only model does not improve AIC; at $n = 12$, mean $\log \text{Var}[g]$ is -5.85 for both $L = 1$ and $L = 3$.

The corresponding decay is approximately

$$\text{Var}[g] \sim n^{-2.15}.$$

For comparison, forcing a linear-in- n form gives a slope of -0.261 with cluster-bootstrap 95% CI $(-0.311, -0.206)$, excluding the Haar-random rate $-\ln 2 = -0.693$. Mean $\log \text{Var}[g]$ falls from -3.55 at $n = 4$ to -6.47 at $n = 14$ for $L = 1$.

These observations do not contradict asymptotic barren-plateau results for deep, sufficiently random

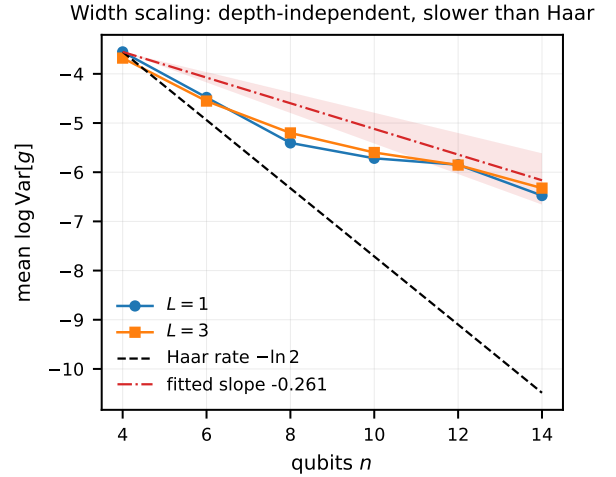


Figure 3: Mean $\log \text{Var}[g]$ against circuit width, by depth. The measured decay is substantially slower than the Haar-random reference.

circuits. They instead show that the finite-width regime tested here is not described by the particular nd phenomenological form used in H1. **For shallow circuits with local observables and encoded real data, the leading dependence can differ from the asymptotic Haar-random scaling.**

6.7 Refitting the complexity correction on the corrected baseline

The exploratory width-scaling result raises a direct question: **does the complexity correction become predictive once the leading width dependence is corrected?** We therefore refit the correction on the n -scaling arm, using the $1 + \log n$ baseline. This analysis is exploratory (Table 10).

Adding $\kappa^\top \mathbf{v}$ to $1 + \log n$ improves AIC by 492.5. The apparent in-sample strength is misleading because $\mathbf{v}$ is a dataset-level covariate: the nine-dataset arm contains only nine distinct values of $\mathbf{v}$ across 12,960 records. AIC therefore counts thousands of repeated circuit observations as independent evidence for a predictor whose effective sample size is nine

Table 11: Quantum test error does not unpin with circuit width. All four arms use the same nine-dataset subset and the same eight non-reuploading encodings, so the rows are like-for-like; $n = 8$ is the frozen grid restricted the same way. *Exploratory*: these data were collected after the fixed-width result was known and are reported in the register of Sec. 6.6, not as a confirmatory test. The pre-declared reading was that a_q counts as unpinned if the across-dataset spread rises above about 0.05 with the near-chance fraction falling materially below 0.8; no width approaches either.

width	cells	mean a_q	sd across datasets	fraction $a_q \geq 0.45$
$n = 4$	2160	0.492	0.014	0.850
$n = 6$	2160	0.493	0.016	0.856
$n = 8$	2160	0.484	0.019	0.827
$n = 10$	2160	0.484	0.019	0.825

datasets.

The consequence is visible under LODO validation. The LODO R^2 falls from +0.158 for the $1 + \log n$ baseline to -32.9 with the full complexity vector. A one-component model gives only +0.0115 and the best pair +0.0143, both selected post hoc. Randomly re-assigned complexity vectors give a null best-of-nine gain with mean +0.0067 and 95th percentile +0.0224, placing the observed one-component gain at $p = 0.235$.

Thus the H1 null is not explained solely by the original width baseline. Even after correcting the leading scaling form, the complexity vector does not add robust held-out predictive information in this nine-dataset arm. More generally, dataset-level predictors are constrained by the number of independent datasets, not by the number of circuit and seed observations.

6.8 Two axes that carry no variance in a_q

The degeneracy of Sec. 6.5 is the paper’s central mechanism, and the obvious objection to it is that $n = 8$ was a pre-specified choice made before anything was known. If quantum test error were width-dependent, the whole H5 analysis would be a statement about one circuit size. We therefore repeated the Phase B protocol at $n \in \{4, 6, 10\}$ on the nine-dataset subset, restricted to the eight non-reuploading encodings, and recomputed the $n = 8$ baseline the same way so the arms are comparable (6,480 additional cells). This arm is *exploratory* in the specification’s sense, it was collected after the fixed-width result was known.

The pre-specified reading was that a_q counts as unpinned if the across-dataset spread rises above about 0.05 with the near-chance fraction falling materially below 0.8. **No width approaches either threshold.** The spread is 0.014–0.019 throughout and $n = 10$ lands exactly on the $n = 8$ baseline; the near-chance fraction never drops below 0.825. Quantum test error is pinned at every width tested.

This matters more in combination than alone. We

already reported that the *encoding* axis collapses: mean a_q spans only 0.462 to 0.497 across all ten encodings, a range smaller than the within-encoding spread, with no encoding winning systematically. The width sweep adds a second, independent axis with the same outcome. **Neither the representation chosen nor the circuit size moves a_q off chance on this suite.**

We also checked a third candidate explanation and closed it. Appel (2026) shows that angle-encoded quantum kernels collapse once the feature map is *wider than the intrinsic dimension of the data*, with the correlation fractal dimension D_2 giving the threshold. Several suite datasets are low-dimensional manifolds in higher ambient space (**synth-2sphere** measures $D_2 = 2.00$, **synth-torus** 2.14), so at $n = 8$ they are encoded three to four times wider than their intrinsic dimension. But the suite is not uniformly over-encoded, **real-higgs** ($D_2 = 10.08$) and **adv-encoded-random** (11.03) are *under*-encoded; and the two groups are indistinguishable in a_q (0.480 over-encoded, 0.492 under-encoded), with $\text{corr}(D_2, \overline{a_q}) = +0.18$ across 27 datasets. Encoding width relative to intrinsic dimension does not explain the pinning either.

It licenses the generalization that the fixed-width design could not support: **the degeneracy is a property of the data and the model class, not of the $n = 8$ choice**, and the H5 verdict is not an artifact of a single circuit size.

It does *not* license the stronger reading that representation is irrelevant. Appel (2026) finds feature-map geometry determines whether a quantum kernel retains any information at all, and Peng et al. (2026) finds a map aligned to the data’s interaction structure beats classical kernels outright. The defensible statement is narrower and more useful: **no representation reachable within this library, at any width tested, rescues a dataset whose structure is already classically accessible.** On the spectral reading these encodings all deliver the same low-degree content to the model, and a quantum model given low-degree structure has nothing a classical model lacks. **That is a claim about the interaction of representation with *this data*, and it predicts its own reversal: change the data so the signal sits only at high degree, and the picture should invert. Part III does exactly that.**

6.9 H3: expressibility–data alignment

The pre-specified primary criterion is satisfied. Across all 27 leave-one-dataset-out folds of the corrected re-run, the fitted slope is $\beta = +0.503$ with a fold-bootstrap 95% CI of $(+0.474, +0.531)$, positive in 27/27 folds. The independent GBM head gives $\beta = +0.395$, CI $(+0.381, +0.411)$. H3 requires $\beta > 0$ with the CI excluding zero, and that criterion is met.

The alignment head has no held-out predictive power: mean gap $R^2 = -0.013$ for the GP and -0.007

for GBM, both worse than the mean predictor. A positive β can therefore reflect the level of the generalization gap without tracking its variation. The observed gap has mean $+0.0042$ and standard deviation 0.0471 ; the fitted relationship can reconstruct this mean without providing useful cross-dataset prediction.

H3’s secondary criterion, Spearman $\rho \geq 0.3$ between predicted alignment and observed negative gap, could not be computed from the stored confirmatory outputs because the per-point alignment scores and observed gaps were not serialized. We record this as an instrumentation failure, not as a null result. The serializer has since been fixed. The negative gap R^2 nevertheless provides direct evidence that the alignment head did not generalize.

Correction of the gap convention. The initial confirmatory run reported $\beta = 0$ on every fold and a gap $R^2 \approx -300$. The discrepancy arose because the implementation treated the already-computed observed gap as though empirical error still had to be subtracted. The corrected implementation returns the gap directly, matching the stored g_{obs} . A synthetic validation recovered the expected relationship with gap $R^2 = +0.86$.

The correction does not alter the H5 conclusion. The GBM edge is bit-identical between the two runs (pooled $R^2 = -0.579$), while the jointly fitted GP edge moves from -0.322 to -0.370 ; both remain far below the H5 threshold of 0.4 .

H3 therefore passes its literal primary criterion but not its intended predictive interpretation. **This illustrates the distinction between a coefficient having the expected sign and a model whose corresponding quantity generalizes across datasets.**

6.10 H4: per-feature encoding optimality

H4 is not tested because it is conditionally downstream of H5. The hypothesis asks whether ansatz configurations that route each feature to its predicted-optimal encoding outperform monolithic encodings. The routing rule itself is produced by the selector. With pooled held-out $R^2 = -0.32$, the selector provides no demonstrated basis for choosing those encodings on unseen datasets.

Running H4 despite this failure would conflate two questions: whether mixed encoding can help at all, and whether the selector can identify when it should help. The former can be tested with an empirical oracle routing, which establishes the achievable ceiling; the latter should only be tested against that ceiling once a valid selector exists. We therefore leave H4 untested under the pre-specified hierarchy.

6.11 H6: dequantization stratification

H6 is not evaluable because the confirmatory suite contains no Mode-Q cells. The access-model condition is nevertheless important: an apparent quantum advantage is not meaningful if the quantum method receives a stronger data-access primitive than its classical comparator. Mode-Q would therefore require a classical or quantum-inspired comparator under the corresponding sampling or query access model. The present suite uses Mode-S or Mode-N assumptions, so H6 remains a framework boundary condition rather than an empirical result.

6.12 A designed quantum-signature control

The H5 null could have two interpretations: either the proposed predictor fails even when a quantum-sensitive signal exists, or the confirmatory suite contains little quantum structure for the predictor to rank. We distinguish these possibilities with a designed quantum-signature control.

We construct a shadow-hard parity probe in which states are labelled by a weight- w global parity and sampled within the corresponding eigenspace. The classical comparator uses the best of the same six models applied to low-order ($k \leq 2$) classical-shadow features. For $w \leq 2$ the edge is approximately zero; for $w \geq 4$ a structure-aware measurement produces an edge of approximately $+0.42$, with $a_q \approx 0$ against $a_c \approx 0.42$ (Fig. 4).

The control separates the existence of a quantum-sensitive signal from the ability of a generic trainable model to discover it. A generic hardware-efficient ansatz with $L = 3$ and a Z_0 readout remains near chance. A trainable measurement-selection head that must discover the parity support also fails for $w \geq 3$. At $w \geq 4$, neither trainable model captures the available $+0.42$ edge.

This control has a deliberately narrow purpose. It shows that the constructed instance contains a quantum-signature structure for which a structure-aware measurement yields an observable edge, while the generic models used here do not learn that structure. It is not a classical-versus-quantum separation result. In particular, the present comparison does not equip a classical method with explicitly constructed higher-order parity features, the same prior knowledge of the planted structure, or a matched access and measurement/query budget.

A stronger parity benchmark would pre-specify these matching choices: the classical feature construction and its order, the structural information supplied to each procedure, the raw-data access model, and a comparable measurement or query budget. Such a study would answer a different question from the present control and is left for future work; we therefore make no claim that the observed edge persists under those stronger classical comparisons.

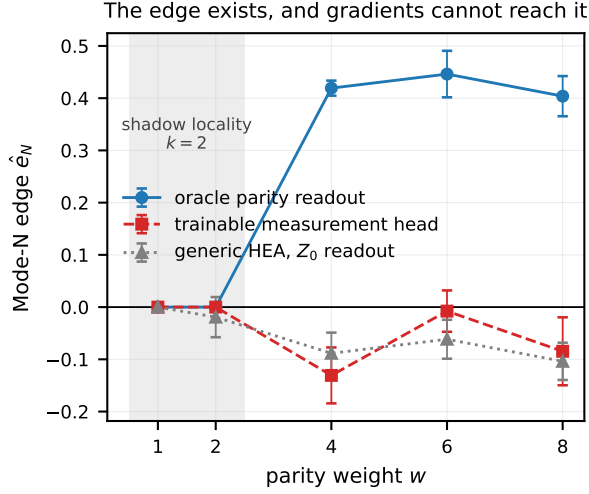


Figure 4: Designed quantum-signature control as a function of parity weight w . The structure-aware measurement turns the edge on once the parity exceeds the $k = 2$ shadow locality, while the generic and trainable readouts remain near chance. The control demonstrates that a baseline-robust quantum-sensitive signal can exist without being discoverable by a generic gradient-trained model.

The result should be read as a control, not as a general theorem about variational quantum algorithms. Here the quantum edge is accessible when the relevant measurement is specified and disappears when the same structure must be inferred by the trainable models. This distinguishes two questions that are otherwise conflated: whether quantum-sensitive structure exists in the data, and whether a generic learning procedure can discover the operation that exploits it.

Taken together, the confirmatory and control experiments support a narrower conclusion than either a universal null or a universal advantage claim. **On the classical benchmark suite, the quantum model is near chance and the measured edge is strongly determined by the classical comparator. A baseline-robust edge can nevertheless be constructed when the data contain a deliberately inaccessible global structure. The central limitation is therefore not the logical possibility of a quantum edge, but the absence of a robust, learnable quantum quantity for the proposed data-complexity selector to predict in the tested regime.**

7 Discussion

7.1 Analysis of Part I approach

This part is intended to provide information about what Part I proved to be correct and what was discarded.

What survived. *The quantities are real and measurable.* All nine complexity components were esti-

mated across the broader 32-dataset frozen registry with reported uncertainties, and the estimators were validated against closed-form testbeds before any experiment. The confirmatory predictor, however, was evaluated on the 27-dataset A/C/D suite defined in Sec. 5. Part I’s operational content, that one can put a number on the structural, statistical, algorithmic and topological richness of a dataset, holds up.

The framework was falsifiable, and that is not a small thing. Part I made claims sharp enough to be killed by an experiment, which is why this paper exists. Most systematic re-benchmarking repeatedly finds that reported advantages do not survive a tuned classical comparator (Bowles et al., 2024; Yu et al., 2026; Leither et al., 2026), which is only diagnosable because the original claims were rarely stated as something that could fail.

The embedding-dependence observation was correct. Part I noted that “the perceived complexity of the same classical dataset is not intrinsic, but depends on the choice of feature map.” That observation is right, and it is what motivates the $\mathbf{v}/\mathbf{q}$ separation of Sec. 3; only its formalization as a property of D was mistaken.

The direction of the data-modulation claim has support. Two of nine complexity sensitivities are significantly non-zero with the sign Part I predicted, higher-order mutual information and MDL compression length, both with $\kappa_i > 0$, meaning higher complexity implies a stronger plateau. Part I’s intuition that data modulates trainability is not empty.

What did not survive. *The scalar composite* (Part I Eq. 7) is dimensionally incoherent and unidentifiable, and *we withdrew it before testing it* (Sec. 3). The same applies to the generalization model of Part I Eq. 15, whose absolute difference between an unbounded KL divergence and a normalized scalar was not a testable quantity. These were corrected by preparing the experiment, not by running it.

The data-gated barren plateau (Part I Eq. 9) fails twice over. It fails on theory, by implying the absence of plateaus on trivial data, contradicting the circuit-ensemble result. And the corrected form of Eq. (7) then fails on measurement: over $n = 4$ to 14 the decay is power-law in width, independent of depth, and slower than Haar-random (Sec. 6.6). Part I’s error here was not the direction but the functional form, and the revision inherited the error.

The topology pillar is untested, not refuted. Part I gave topological complexity a load-bearing role, extended in its Eq. 16. Our pre-specified test appeared to refute it; adding C_{top} to the barren-plateau model changed AIC by 0.1, but the estimator behind that number was measuring dispersion and ambient scale rather than topology, and we withdraw the result (Sec. 6.4, Appendix A). On a corrected and shape-validated estimator, the answer is genuinely ambigu-

ous: threshold-free summaries still show nothing, a significance-thresholded bar count clears the AIC criterion, and which of these is the right summary is not something our design can adjudicate after the fact. The curvature pillar added in Sec. 3 is untested here by construction. So “geometric structure matters” remains *open* in both halves, which is a weaker and more honest position than the one we held before the audit, and it is the single largest piece of unfinished work this paper leaves behind.

We flag it for a *dedicated* successor rather than for the next paper in this series, because the two halves fail for different reasons and neither is repaired by more of what this suite contains. The topological half is a **specification** problem: on the corrected estimator the verdict depends on which summary is chosen, threshold-free or significance-thresholded, and that choice must be fixed in advance, it cannot be adjudicated post hoc by us or by anyone reading these data. The curvature half is a **data** problem: v_5 needs graph- or hypergraph-native inputs, which a point-cloud suite structurally cannot supply, so no reanalysis of the frozen grid will ever exercise it. A study that fixes both needs a different instrument and a different corpus from the one built here, which is why we do not fold it into the roadmap below as an increment.

It is nonetheless committed work rather than a gesture: the program this paper belongs to is developing it as a dedicated geometry-and-topology study, carrying forward the corrected estimator and the shape validation established here (Appendix A) and adding the graph-native corpus this suite lacks. Whether it appears as a further part of this series or as a standalone paper is not yet fixed, and we prefer to leave that open rather than promise a number we may not keep: the sequence from Part III onward is driven by which instrument the preceding result demands, not by a plan settled in advance.

The central claim fails its pre-specified test, and the reason bounds what that failure means. Complexity does not rank the edge: pooled leave-one-dataset-out $R^2 = -0.32$ against a pre-specified 0.4, with the fold-bootstrap interval entirely below threshold. But the quantum model is at chance on most of the suite (Sec. 6.5), so **on these datasets there was no quantum signal for complexity to rank**. The result is therefore a statement about a regime, not about the framework in general.

The pattern, stated plainly. Part I’s *observations* were largely sound; its *functional forms* were not. Every specific equation proposing how complexity enters, the linear composite, the exponential data-gated plateau, the absolute-difference alignment, was either dimensionally incoherent, contradicted by established theory, or empirically unsupported, while the underlying observations that motivated them survive. The framework’s failure mode was that its quan-

ties were real and its equations were guessed.

There is a methodological corollary worth recording. Three of the six issues in Sec. 3 were found by *implementing* the framework, before any data was collected: dimensional analysis of the composite, correlation structure among components, and the data-free limit. Implementation is a form of theory criticism.

7.2 The four lessons, as recommendations

Each lesson converts into a concrete requirement on future work.

Lesson 1 $\Rightarrow$ report a baseline sweep, not a baseline. Any advantage measured against a single classical model should be treated as unquantified until the baseline is varied. We recommend reporting the edge against a *declared baseline sweep*, including a strong comparator such as the best of a fixed standard set, and publishing the per-baseline breakdown, so that a reader can see how much of the claimed edge is model-selection slack. **Had we not varied the baseline, this paper would have reported a positive, pre-specified, reproducible, and spurious result.**

Lesson 2 $\Rightarrow$ separate the existence of an edge from its reachability. Our positive control shows these come apart cleanly: the edge is present at the level of the measurement but absent from the gradient-trained models tested here. **A selector trained on gradient-trained outcomes therefore has a target that is systematically missing exactly the advantages worth having.** Advantage claims should state which of the two they concern.

Lesson 3 $\Rightarrow$ vary circuit width, and fit the regime you are in. Benchmark suites at a single width cannot test the functional form of any scaling law in n ; ours could not, until we added an arm for it. Doing so overturned the form: decay is power-law in width, depth-independent, and slower than Haar-random. **Any framework predicting advantage from circuit-scaling arguments should be fit in the finite regime rather than extrapolated from asymptotic results**, and should report the width sweep that licenses its form.

Lesson 4 $\Rightarrow$ check that your axes carry variance before fitting a model on them. A framework is only tested on the axes along which its target actually varies. Ours did not vary: the quantum model sat at chance in 81.5% of cells, so the edge collapsed onto classical error ($\rho(\hat{e}, a_c) = +0.913$), and the encoding axis carried 0.5% of the variance against 83.8% between datasets. The consequence is that $\mathbf{q}(D, \phi)$ was never given a chance to matter, and the null on $\mathbf{v}(D)$ is a statement about a regime rather than about complexity. Before fitting a predictor, report the variance

decomposition of the target over every axis the model uses, and it distinguishes a hypothesis that failed from one that could not have been tested.

And a corollary on where to look next. The only baseline-robust edge we found was engineered to be non-local in the measured observables, and its onset tracked that locality precisely. This suggests the informative axis is the locality structure of the label-observable relationship, not aggregate data complexity, a sharper and more testable target than the one Part I proposed.

7.3 Another interpretation, and what it would change

The results above are reported as a falsification, and that reading stands: the selector does not predict the edge, and none of the hypothesis outcomes is rescued by a more careful analysis. But there is a second reading of the same evidence which does not weaken the null, and which points somewhere different. We set it out here rather than in the conclusion, because it is an interpretation rather than a result.

The quantity we chose to predict is a comparison. The primary edge

$$e_{\text{budget}} = a_q^{\text{budget}} - a_{\text{LR}}$$

is, by construction, a difference between two rival systems. Section 6.2 shows what follows: widening the classical model set moved the selector’s leave-one-dataset-out R^2 from +0.13 to −0.94 without changing a single quantum measurement. The predictand was not a property of the data, or of the circuit, or even of the pair, it was a property of the comparison, and the comparison moved.

The six-model comparison is a fixed-library sensitivity analysis, not a search for the best attainable classical benchmark. Its purpose is to test whether the apparent selector signal survives a broader, standard set of classical references under the specified protocol. It does not establish the result for a hyperparameter-tuned classical frontier. Conversely, because no modest tuning study was run, we do not infer the direction or size of any further change in the measured edge.

A useful follow-up would pre-specify a modest and equal tuning budget for the classical library and report it as an additional sensitivity analysis, without redefining the pre-specified LR-based H5 target.

That is worth separating from the framework’s content. **A target defined against a baseline can be destroyed by improving the baseline, and no amount of better feature engineering on the quantum side prevents it. Whether data complexity predicts something is a different question from whether it predicts this something.**

Another interpretation of the designed quantum-signature control. We reported in Sec. 6.12 that the only baseline-robust edge we found was unreachable by gradient-based training: the advantage sat at the level of the measurement, not the optimization. Read as a competition result, that is a quantum win that cannot be trained for, and therefore of limited use.

Read structurally, it says something else. It says the two substrates were good at different halves of the same problem, **the quantum device at preparing and measuring a state whose label is invisible to low-order classical features, the classical optimizer at everything downstream of that measurement.** The failure was in requiring one substrate to do both. On that reading the designed control is not a curiosity attached to a null result but a worked example of a decomposition, and **the natural follow-up is not a better selector but a better division of labour.**

Some changes need to be proceed to test this interpretation. A target internal to a single pipeline rather than defined against a rival, for instance the marginal contribution of a quantum component, measured by replacing it with a capacity-matched classical control and holding the rest of the workflow fixed. Such a quantity cannot be moved by widening a baseline set, because there is no second system in the comparison. We do not test it here, and we note that it carries its own failure mode: **a marginal-contribution measure is only as strong as the control it is measured against,** which is Sec. 6.2’s lesson in a new place rather than an escape from it. Sec. 7.5 sketches what a study built on this footing would look like.

7.4 Limitations

A falsification is only valuable if the case that the null is a fact about the framework rather than an artifact of the setup. We therefore separate the limitations by what they threaten.

Here are the limitations that do not plausibly flip the verdict. $N = 27$ datasets is small for leave-one-dataset-out generalization, and it does limit effect-size precision: we can say the selector carries no held-out signal, not how much signal a better one would carry. But the pre-specified threshold is $R^2 \geq 0.4$ and the measured value is −0.32 pooled, with the GBM head at −0.58, worse than predicting the mean, not marginally short of a bar. Recovering H5 from this sample would require the selector to have been unlucky in a consistent direction across 27 independent folds. The same applies to H1, where the observed $R^2 = +0.026$ sits against a pre-specified 0.7. Both verdicts are also invariant to the β -convention correction of Sec. 6: the GBM edge path is bit-identical across runs and the GP edge moves only from −0.322 to −0.370.

Now, the limitations that bound what the null cov-

ers. These do not weaken the result but do restrict its scope, and each is a concrete way the framework could survive in a regime we did not test.

- *Circuit regime.* All pre-specified circuits are $n = 8$ qubits, simulated, noiseless, NISQ-scoped and fault-tolerance-free. Width dependence is no longer untested: gradient variance was measured at $n \in \{4, \dots, 14\}$ (Sec. 6.6) and *quantum test error* at $n \in \{4, 6, 8, 10\}$ (Sec. 6.8), the latter closing a gap the earlier arm left open, the scaling sweep measured gradients only, so a_q had been untested at *any* width other than 8. Both are exploratory. Behaviour above $n = 14$ remains untested, as does hardware noise.
- *The classical software layer is a possible and undiagnosed source of loss.* Every number here comes from an 8,100-cell simulator sweep, and Panambalom et al. (2026) show that postprocessing routines developed against simulators can encode assumptions that fail at device scale, producing data loss *not detectable from high-level model outputs*. Our pipeline is simulator-native throughout, so the device-scale form of that failure does not arise; but the general point, that a plumbing defect can be invisible in the outputs one inspects, is one this paper has already been bitten by once, in the C_{top} estimator of Appendix A. We have not audited the postprocessing layer with the same care we eventually applied to that estimator, and we do not claim to have done so.
- *Encoding and ansatz libraries are finite.* Four encoding families in ten configurations, and a hardware-efficient ansatz family. A selector that fails over this library might succeed over another. Re-uploading encodings are excluded from H1/H2 outright, because they admit no fixed encoded state to compute $\mathbf{q}$ on.
- *The fifth pillar was never exercised.* Every dataset in the frozen suite is non-graph, so the curvature component v_5 is dropped throughout and H2 reduces to a test of the topological contribution alone. Curvature is scoped out, not falsified.
- *Mode Q is empty.* No encoding in the suite assumes coherent QRAM loading, so H6 has no stratum to test and the framework’s claim to discriminate dequantizable from non-dequantizable regimes is unexamined.
- *The two tracks are parallel, not unified.* The designed quantum-native control (Sec. 6.12) is a positive control, not an input to the selector; nothing here shows what a selector trained on quantum-native data would do.

- *The target variable is cell-dependent, and we tested only one.* Sec. 7.5 argues the framework’s claim is properly stated per (problem type, algorithm, data-access mode) cell. The same applies to the *predictand*: $\hat{e}$ compares a quantum method against a classical rival, which presumes the two are substitutes. In cells where they are complementary rather than rival, a competitive target may be the wrong instrument regardless of whether data complexity is the right axis (Sec. 7.3). The result of this study is a null for this target in this cell.

The embedding-interaction half was never given a chance to matter. The framework has two halves, intrinsic complexity $\mathbf{v}(D)$ and embedding interaction $\mathbf{q}(D, \phi)$, and only the first was tested against a target with structure to predict. Decomposing the variance of the budget-matched edge over all 8,100 cells: 83.8% **lies between datasets**, 0.5% **between encodings** and 0.5% **between ansätze**. So $\mathbf{v}(D)$ faced a real target, while $\mathbf{q}(D, \phi)$ faced an axis carrying half a percent of the variation.

The cause is the degeneracy of Sec. 6.5 acting on a second axis. Because no configuration escapes chance, mean a_q spans only 0.462 to 0.497 across all ten encodings, a range *smaller* than the within-encoding spread of 0.05–0.09, the encoding dimension collapses. Nothing distinguishes a product-state angle encoding from a ZZ -map producing genuinely entangled states (mean half-cut entropy 1.52 nats of a possible 2.77), because both land at chance. Consistently, no encoding wins systematically: across the 27 datasets the best-performing encoding is scattered over nine of the ten configurations, as expected from noise.

This is not a defect in any encoding. It is that an interaction term cannot be tested when the quantity it modulates is pinned. **We therefore report $\mathbf{q}(D, \phi)$ as untested rather than unsupported**, and note that any future design must establish that its encoding axis carries variance *before* fitting a model that depends on it.

We also report limitations in the instruments themselves. Three of the nine complexity components are measured more weakly than the framework assumes. Topological metrics are relative rankings across the suite, not absolute Betti detectors, so C_{top} orders datasets but does not characterize them, and in the form used for the pre-specified tests it did not even order them by topology (Appendix A). The corrected estimator is scale-invariant and shape-validated, but it normalizes by the global diameter, so widely separated clusters dilute the measure and multiplicity is lost: three well-separated circles score *below* one. Local-scale normalization repairs that at the cost of sample-size invariance, which matters more when comparing datasets of differing n ; we took the trade deliberately and record it as a limitation of the corrected instrument. Higher-order mutual-information

estimation is brittle at $k \geq 3$, which is precisely where the correlation-order pillar should carry its distinctive content, so that pillar is weakest exactly where it matters most. And the H3 secondary criterion, specified to catch the failure mode we ultimately diagnosed by other means, was lost to a serialization defect (Sec. 6.9); we report the gap R^2 as a proxy and the serializer is fixed, but the pre-specified statistic itself is unavailable for these runs.

The latest is the limitation that cuts against our own positive result. Lesson 1 (Sec. 6.2) says the classical baseline determines the measured edge, and that applies to the designed quantum-signature control as much as to anything else. The control edge is measured against the best of six models on $k \leq 2$ shadow features; it is baseline-robust with respect to *that* set, and the construction gives a principled reason to expect robustness. The label is provably invisible to any $k < w$ Pauli feature, but a classical learner given higher-order shadow features, or a different representation entirely, has not been ruled out empirically. We report the edge as surviving the sweep we ran, not as a proof that no classical method reaches it.

The parity control also has a limited comparative scope (Sec. 6.12). Its positive result establishes recoverable signal under the specified structure-aware measurement; it does not benchmark against classical procedures supplied with matched higher-order features, prior structural knowledge, or a matched measurement/query budget.

7.5 Scope of the algorithm suite, and the roadmap to Part III

The validation instantiates a *single* quantum-algorithm family: parameterized-circuit classification, comprising an embedding ϕ , a variational ansatz $U(\theta)$ and a single-Pauli $\langle Z_0 \rangle$ readout, trained by parameter-shift descent. Every task is supervised binary classification on classical feature vectors, which places all pairs in Mode S. The selector is defined for any (algorithm, mode) cell, but only this one cell is populated.

The primary result must therefore be read conditionally. What H5 tests is “data complexity predicts the edge of variational classification on Mode-S data”, not a claim about quantum machine learning at large. The scoping is not incidental: Mode-S supervised learning on unstructured classical vectors is precisely where classical surrogates are strongest, and the primary comparator is the pre-specified logistic-regression baseline, supplemented by a fixed six-model sensitivity analysis. The null therefore applies to this specific Mode-S variational-classification comparison, not to quantum machine learning at large.

The two cells where separation is theoretically most defensible, **Mode N** and **Mode-Q sparse-survivor**, are exactly the ones this suite does not populate (Ta-

Table 12: The honest unit of the framework’s claim is a (problem type, algorithm, data-access mode) cell. Part II populates one row.

Problem type	Algorithm	Mode	Separation status
Supervised classification (<i>this work</i>)	variational / kernel	S	weakest; kernels often dequantizable
Combinatorial optimization	QAOA, annealing	S	contested vs. classical heuristics
Linear systems, PCA	HHL, QSVT	Q	dequantizable unless sparse-survivor
Ground states, dynamics	VQE, Trotter	N	quantum-native; separation depends on problem family
Generative modeling	Born machine	S/N	partial

ble 12). Extending to a full algorithm $\times$ mode matrix is the principal open direction, and the quantum-native cell, for which this work already supplies an honest classical-shadow baseline and a positive control, is the natural first expansion. Part III develops it, on the quantum-native probe ensembles drawn from this corpus; it does not carry the mode taxonomy of Table 12 across, so the correspondence is by construction rather than by name.

Also outstanding, in the order we would rank them. **First, geometry:** the topological verdict is summary-dependent and the curvature pillar was never exercised (Sec. 7.1), and of everything listed here it is the item whose resolution would most change the framework, since on this suite the composite *is* the topology component ($\rho = 0.998$). It needs a pre-specified topological summary and a graph-native corpus, so it is a separate study rather than an extension of this one, and one the program is developing (Sec. 7.1). Then: H4, which needs the mixed-encoding implementation; an n -scaling arm promoted from exploratory to pre-specified; a QRAM-dependent encoding to make H6 evaluable; and hardware.

8 Conclusion

We set out to test whether data complexity predicts quantum machine learning advantage, under pre-specification, and it does not. On a frozen suite of 27 classical-track datasets, 10 encodings, and 6 ansätze, a selector trained on the complexity and interaction vectors fails to predict the budget-matched edge on held-out datasets: pooled leave-one-dataset-out $R^2 = -0.32$ against a pre-specified threshold of 0.4, reproduced by an independent gradient-boosted head. The barren-plateau scaling law fares no better;

the data-modulation term is detectable and correctly signed for two of nine components but explains almost none of the variance, and persistent homology contributes nothing. Under the pre-specification, this rejects the framework’s central predictive claim in the tested regime.

The reason the result is worth reporting is that we can say *why*, and each reason outlives the framework that produced it. The apparent complexity–advantage signal was real but misattributed: it tracked where a linear baseline fails, and a single weak baseline is enough to manufacture a fully predictable, reproducible and spurious quantum advantage. Where an edge did survive a best-of-six classical comparator, a shadow-hard parity construction, at +0.42, it proved unreachable by gradient training, in a way that unifies classical parity hardness, barren plateaus and shadow-hardness as one absence of low-order signal. And varying circuit width, which a fixed-width benchmark cannot do, showed the scaling law to be misspecified in the regime experiments actually occupy: power-law rather than exponential, depth-independent, and slower than the Haar-random rate. A selector trained on gradient-trained outcomes was therefore fitting a small correction on top of a wrong leading term, to a target that systematically excludes the advantages worth having.

What we take from this is narrower and more useful than the hypothesis we started with. Aggregate data complexity is the wrong axis; the locality structure of the label–observable relationship is a sharper one, and it is what the only surviving edge was built from. The claim is also properly conditional: one algorithm family, one data-access mode, simulated circuits of at most 14 qubits, and the cells where a separation is most defensible are exactly the ones this suite does not populate. Part III takes up the quantum-native of those. In the meantime we would put the methodological result first: report a baseline sweep rather than a baseline, distinguish an edge that exists from an edge that can be found, and vary circuit width before fitting anything that scales with it. Had we done only the first of those, this paper would have reported a positive result, and it would have been wrong.

In the grey zone

It is worth being exact about the reach of the negative result, because the tempting overstatement is close by and we do not make it. We tested one task formulation, twenty-seven datasets, ten encodings, six ansätze and one data-access mode, and on those we found no advantage attributable to the quantum model. That is not evidence that quantum machine learning over classical data cannot work. We have not tested all classical datasets, and we have no theorem that forbids it; Propositions 1 and 2 constrain what an encoding can *do*, not what a model can *achieve*. An encod-

ing cannot manufacture label information and cannot place preprocessing gains beyond classical reach; neither statement closes the residual term, and the residual is where an advantage would live.

The 27 datasets support a pre-specified test within this benchmark suite, not a population-level claim about all tabular learning problems or all QML pipelines. Leave-one-dataset-out evaluation tests transfer among the sampled datasets, but the suite is not a probability sample of task families, data-generating mechanisms, encodings, or classical competitors. The central conclusion should therefore be read as conditional on the tested regime: the specified complexity representation does not reliably predict the measured edge across these 27 datasets.

What the propositions do supply is a reason the null was not an accident of our choices. The preprocessing-attributable term is classical by construction, so it was never ours to claim. The residual was measured across ten encodings, including entangling maps carrying 1.52 nats of half-cut entropy, and found indistinguishable from zero *in a regime where the quantum model sits at chance in 81.5% of cells*. A pinned quantity cannot exhibit an interaction. That is a statement about the regime we could reach, and the honest form of our claim is conditional on it.

The question we would leave behind

If one thing from this paper outlives the framework it was written to test, we would rather it were a question than a result. Every reported quantum–classical gap decomposes exhaustively, by Eq. (33), into a preprocessing-attributable term and a residual. The first is obtainable by any classical party at the cost of running the preprocessing. Only the second is a quantum claim.

So: **which term did you measure?**

It applies to work this paper never touched; it can be answered from a comparison that is already published, and a study that cannot answer it has not isolated a quantum effect, whatever its headline number. Had we been asked that question at the outset, this paper would have been much shorter.

Data and code availability

All code, specifications, frozen dataset registry, logged deviations and raw results are released at https://github.com/Christophe-pere/data_complexity_framework, made public at publication.

Environment. Python 3.12 with `numpy` 1.26.4, `scipy` 1.13.1, `scikit-learn` 1.6.1, `torch` 2.7.1, `gpytorch` 1.15.2, `lightgbm` 4.6.0, `ripser` 0.6.15, `gudhi` for the pre-specified topological component. Quantum simulation is exact state-vector in `numpy`; no quantum hardware was used.

A Audit and correction of the topological component

The pre-specified H2 test (Sec. 6.4) returned a decisive null. Post-hoc inspection of the estimator behind it showed the null was not interpretable, and this appendix documents what was wrong, what replaced it, and what changes as a result. We report it in full because the defect survived into a confirmatory analysis, and the way it survived is instructive.

A.1 Four compounding defects

The frozen component C_{top} was total Vietoris–Rips persistence, computed with `gudhi`. Four things were wrong with it at once.

1. **H_0 entered at full weight.** The deployed call used `max_dim=2` with default weights $(1, 1, 1)$, so dimension-0 persistence was summed into the composite. H_0 total persistence is n bars each of order an inter-point spacing, a *dispersion* statistic. It carried between 50.7% and 99.6% of the composite across the suite, median $\approx 90\%$.
2. **No scale normalisation.** Persistence was accumulated in absolute length units, so the component measured how large the point cloud is. Over the 27 classical-track datasets, $r(\log \text{diam}, \log C_{\text{top}}) = +0.922$, i.e. $R^2 = 0.85$ (Fig. 5a).
3. **The filtration was truncated too early.** The cutoff was $0.5\times$ the diameter, but genuine features die later than that, a clean circle’s H_1 class dies at $\sqrt{3}r = 0.866\times$ diameter. Real classes therefore never died and were recorded as essential.
4. **Edge collapse deleted the essential classes.** Edge collapse (Boissonnat–Pareek) is persistence-preserving for a *complete* flag filtration, not a truncated one. Defects 3 and 4 compound: truncation manufactures essential classes and collapse removes exactly those. Directly demonstrated on a clean circle, at cutoff 0.5, $n(H_1) = 0$ with collapse and $n(H_1) = 1$ ($P_1 = 0.825$) without; at cutoff 0.9 the two agree exactly.

The consequence is stronger than attenuation. **The deployed estimator returns $P_1 = 0.000$ for a perfect circle:** it removed topology rather than underweighting it.

Why this survived validation. The validation suite exercised C_{top} with `max_dim=1` and weights $(0, 1)$, H_0 explicitly zeroed, while the feature-extraction path used `max_dim=2` with default weights. The validated configuration was never the deployed one. This is the generalizable lesson: validating an estimator is not the same as validating the estimator *as called*, and only the latter licenses a null result.

A.2 The corrected estimator

We rebuilt the component rather than patching it. The backend is `ripser`, which computes the full untruncated Rips persistence directly, so defects 3 and 4 cannot recur by construction. H_0 is excluded; only dimensions ≥ 1 count. Every bar is divided by the point cloud’s diameter, making the measure scale-invariant. Point clouds are subsampled to 250 points with a fixed seed.

Because “how much topology” has no single canonical summary, we report several and treat the choice as an open specification decision rather than settling it after the fact: ℓ_2 (sum of squared normalized persistences, threshold-free), the longest single bar, and a count of bars exceeding a significance cut θ , together with their total persistence.

A.3 Validation against known shapes

We validated against point clouds with known Betti numbers (circle, $\beta_1 = 1$; torus, $\beta_1 = 2$, $\beta_2 = 1$; 2-sphere, $\beta_2 = 1$) and topologically trivial controls (a 3D Gaussian blob and a uniform box). A usable summary must separate the two groups.

The pre-specified form fails outright (Table 13, Fig. 5c), and its failure mode explains the downstream damage: total persistence rewards bar *count*, so a 3D Gaussian blob’s hundreds of tiny sampling-noise cycles outscore a clean circle’s single long bar. Every corrected summary passes, with and without Mahalanobis whitening.

The corrected component is invariant where it must be. On a circle with the ℓ_2 summary it is exact under rescaling ($r = 10^{-2}$ to 10^3 : flat at 0.628) and under ambient-dimension padding ($d = 2$ to 100 : flat), and stable across sample size ($n = 150$ to 1200 : 0.611–0.648) and subsample seed (0.601–0.654). Added noise degrades it smoothly and monotonically ($\sigma/r = 0 \rightarrow 0.8$: 0.628 $\rightarrow$ 0.011). The scale artifact is gone: $R^2(\log \text{diam}, \cdot)$ falls from 0.85 to 0.096 (Fig. 5b).

Two sanity checks on the suite itself. The corrected ranking is topped by the parity and spin-glass families, binary data on hypercube vertices, which genuinely carries rich Rips topology, together with the designed topological probes; the pre-specified ranking was topped by `real-credit-default`, which is simply the largest point cloud (Fig. 5d). And the four parity datasets share an identical X distribution by construction, differing only in their labels: they score 5.98, 5.12, 6.07 and 5.41, equal within seed noise, as they must.

A limitation of the correction. Diameter normalization is a *global* scale, so widely separated clusters inflate the denominator and multiplicity is lost: three circles separated by six radii score *below* one

Table 13: Known-shape validation. A usable topological summary must place every shape with nontrivial homology above every trivial control.

Summary	min(topological)	max(trivial)	separates
<i>total</i> (the pre-specified form)	0.916	2.602	no
longest bar	0.318	0.092	yes
ℓ_2	0.238	0.107	yes
count, $\theta = 0.10$	1.000	0.000	yes
thresholded total, $\theta = 0.10$	0.792	0.000	yes

circle. Normalizing by a local scale (median k -NN distance) repairs multiplicity but forfeits sample-size invariance, which matters more when comparing datasets of differing n . We took the trade deliberately.

A.4 Estimation of what changed

H2 becomes inconclusive (Table 9). The threshold-free summaries still show nothing; a significance-thresholded bar count clears the AIC criterion, but only for $\theta \in [0.08, 0.10]$, failing on either side. A binary presence-of-topology indicator gives $\Delta\text{AIC} = +2.8$, also a fail. We decline to select among these after seeing the outcomes, so the reported verdict is that the question is open.

Accordingly, this result should not be read as a universal negative claim about topology-based predictors. It is a robustness statement limited to the alternative estimators examined here: those alternatives do not recover a stable topology contribution on the present dataset suite.

H1 is unaffected. Pooled LODO R^2 stays within $+0.026$ to $+0.031$ across every variant, against a threshold of 0.70.

H5 is unaffected, and provably insensitive to this component. The obvious objection to a falsification obtained with a broken feature is that a correct feature might have rescued it. We tested that directly rather than arguing it, by varying *only* the topological slot and leaving the other eight components and the target untouched (Table 14).

Every treatment fails, and the informative comparison is the spread: **the gap between a correctly measured component and one replaced by random noise (-1.20 vs -1.25) is far smaller than the gap to the decision threshold (-1.20 vs $+0.40$)**. Deleting the component outright changes little, and the best result of the six comes from *destroying* the signal by permutation, so the slot contributes nothing positive under any treatment. No topological estimator, however well built, could have moved H5 across its threshold.

This is what should be expected once the mechanism behind the H5 failure is identified, because that mechanism sits on the *target* side rather than the feature side: the edge itself is not a stable quantity, collapsing from a 17% to a 1% quantum win rate when the classical baseline is widened from logistic regres-

sion to a best-of-six set (Sec. 6.1). No feature vector predicts a target that is largely an artifact of baseline choice, and no repair to $\mathbf{v}$ addresses that.

The proxy above is a gradient-boosting stand-in for the confirmatory GP pipeline, used because it permits the six-way comparison cheaply. We additionally re-ran the full pre-specified confirmatory protocol, sparse variational GP, leave-one-dataset-out over all 27 datasets with inner γ cross-validation and the three-step EM, end-to-end on the corrected component. It returns **pooled LODO** $R^2 = -0.451$ (GBM: -0.674), per-fold median -1.44 , and a fold-bootstrap 95% interval of $[-55.3, -8.48]$ whose upper end lies far below the pre-specified threshold of 0.40. **H5 fails on the corrected component**, and fails *more* decisively than on the original (-0.32): repairing the topological estimator moved the primary result away from its threshold, not toward it.

Two features of that re-run are worth recording. The per-fold *mean* is -28.8 against a median of -1.44 , the same near-constant-edge blow-up discussed for the original run; the pooled figure is the honest headline in both cases. And the folds where the selector does worst are precisely the topology probes: **synth-klein-bottle** (-273.6), **synth-3sphere-pc** (-157.1) and **synth-2sphere** (-35.8), the three datasets on which the corrected component differs most from the defective one. Whatever the repaired estimator is now measuring, it does not help the selector where topology is by construction the dominant structure.

Software: the defective class is retained unchanged in the released code so the pre-specified numbers stay reproducible; the corrected estimator is a separate class, and the audit is recorded in the project’s deviations log.

The audit changes the interpretation of the topology result, but its scope should be stated carefully. The recomputation is robust to the alternative estimator constructions, normalizations, and summary choices tested in this appendix. It does not establish robustness to every possible topological estimator or to every representation of the data. Within the tested alternatives, we find no evidence that the topology term provides a stable contribution to the H2 target on this suite.

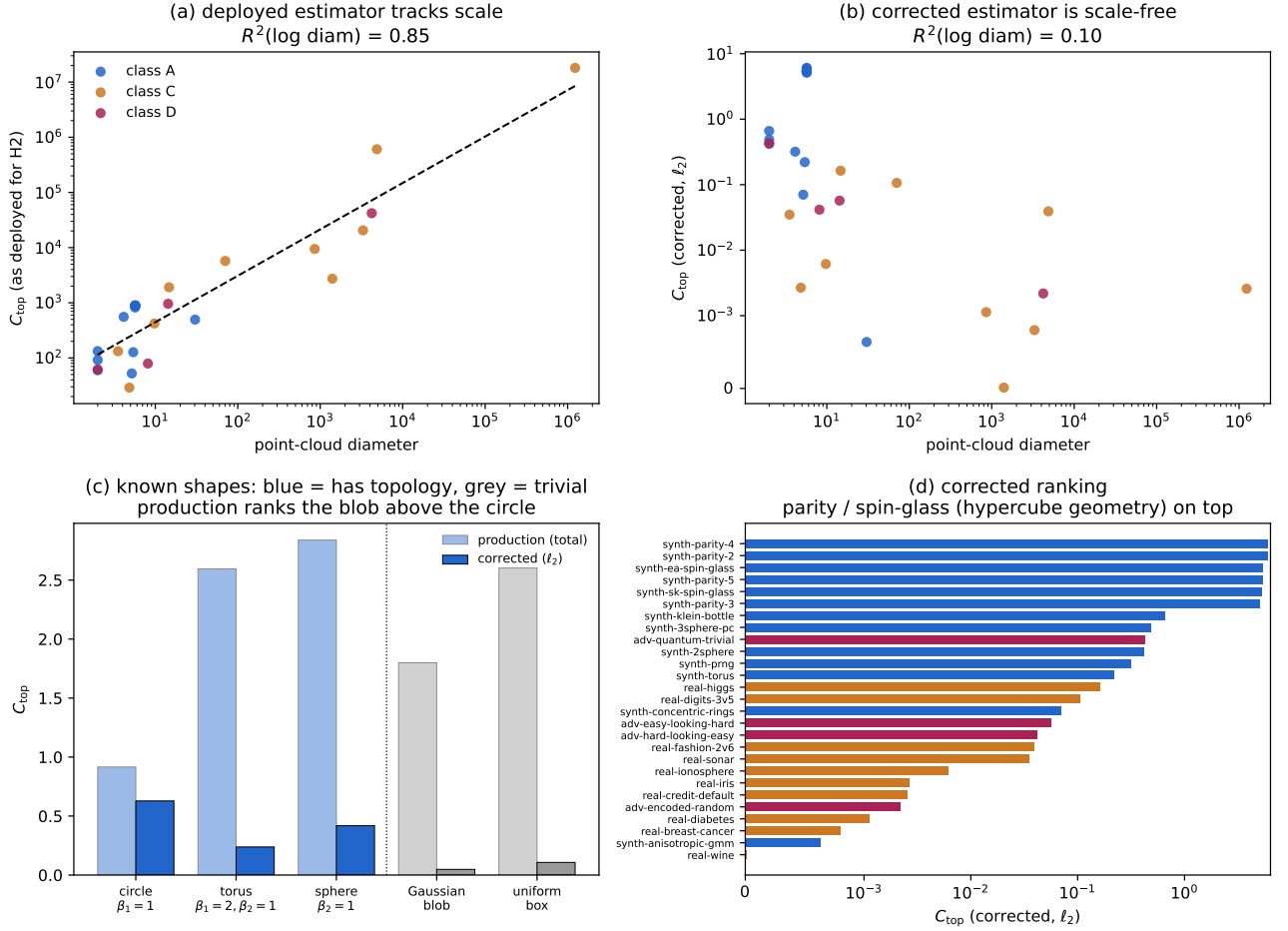


Figure 5: Audit of the topological component. (a) The estimator as deployed for H2 tracks point-cloud diameter across five orders of magnitude, $R^2 = 0.85$. (b) The corrected estimator is scale-free, $R^2 = 0.10$. (c) Known-shape validation: blue bars have nontrivial homology, grey are trivial. The pre-specified form ranks the Gaussian blob above the circle; the corrected form does not. (d) The corrected ranking over the 27 classical-track datasets, coloured by dataset class.

B Per-dataset complexity vectors

Table 15 gives the raw estimator outputs for the nine implemented components of $\mathbf{v}(D)$, alongside the corrected topological component T^* of Appendix A. Table 16 maps these onto Part I’s four pillars and evaluates Part I’s composite C_{data} (Eq. (1)) for each dataset, under both the deployed and the corrected topological estimator. The composite is reported so that Part I’s quantities can be reproduced and compared directly; it is *not* used anywhere in Part II’s analysis, for the reasons measured in Table 1.

Two further Part I quantities have no entry here, in both cases because they are not properties of a classical dataset. $C_{\text{quant}}(\mathcal{E})$ of Part I Eq. 10 is defined on a *quantum ensemble* $\mathcal{E} = \{(p_i, \rho_i)\}_{i=1}^N$ on n qubits, combining entanglement entropy, multipartite information, embedded-kernel effective rank, a magic/nonclassicality monotone $N(\rho)$, quantum Fisher information $F = \text{Tr}[\rho L^2]$ and a quantum topological term. It is therefore a quantity of the quantum-native track, which Part II runs as a parallel arm rather than as part of the classical-data pathway

(Sec. 6.12), and it is not assembled here.

We decline to claim that the Table 1 result transfers to it, because we cannot test that: of Eq. 10’s six components, only three (average bipartite entanglement entropy, multipartite total correlation, and a magic monotone) are implemented in this work, and the quantum-native arm evaluates them on five probe datasets of a single family at fixed width $n = 8$, across which the magic monotone is constant. That design cannot exhibit or exclude variance domination. What can be said is weaker and rests on inspection rather than measurement: Part I declares Eq. 10’s components “normalized, dimensionless” but specifies no normalization for them, and their natural scales are heterogeneous in the same way Eq. (1)’s are; the two entropies are extensive in qubit count; the effective rank $(\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$ is a count on $[1, N]$, the quantum Fisher information $\text{Tr}[L^2]$ is unbounded above and depends on the parametrization, and the quantum topological term is itself a sum of a topological entanglement entropy, an Euler characteristic (a signed integer) and persistent-homology life-

Table 14: Sensitivity of the H5 falsification to the topological component. Only slot 5 of $\mathbf{v}$ varies; everything else is held fixed. Pooled leave-one-dataset-out R^2 on the budget-matched edge over all 8,100 Phase B cells, gradient-boosting predictor. The pre-specified pass threshold is $R^2 \geq 0.40$.

Topological slot	Pooled LODO R^2	H5
as pre-specified (defective)	−1.354	fail
corrected, ℓ_2	−1.201	fail
corrected, count $\theta = 0.10$	−1.148	fail
removed entirely (8 components)	−1.292	fail
replaced by $U(0, 1)$ noise	−1.254	fail
permuted across datasets	−0.616	fail

times (lengths). Whether that heterogeneity actually produces domination is an open question for the quantum-native track, not a result of this paper.

$C_{\text{ind}}(D; \phi)$ of Part I Eq. 11 has no entry either, and this is deliberate rather than an omission. That quantity is not a property of a dataset, so there is no per-dataset column it could occupy; it is replaced by the pair $(\mathbf{v}(D), \mathbf{q}(D, \phi))$ throughout (Sec. 3.3), whose values are the $\mathbf{v}$ columns here together with the embedding-interaction components reported per (D, ϕ) pair in the results.

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AI statement

This study used Claude Opus 5 (Anthropic) as an engineering collaborator, under the author’s direction. The AI was used as a tool to write code and improve text smoothness. All hypotheses, decision rules, direction, reviews, analysis and conclusions were human-made.

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Table 15: Raw estimator outputs for the nine implemented components of $\mathbf{v}(D)$ (plus the corrected topology column T^*), before rank normalization. Reported for reproducibility, and to make concrete the dimensional incoherence motivating Sec. 3.1: the columns differ by up to seven orders of magnitude in range and do not share units. H differential entropy (kNN-KL); d_{pca} , $d_{2\text{nn}}$ intrinsic dimension (PCA-effective, TwoNN); K compression ratio (gzip); T topological complexity as pre-specified (Vietoris-Rips, defective — see Appendix A); T^* its corrected replacement (untruncated Rips, H_0 excluded, diameter-normalised ℓ_2); I_2 , I_h second- and higher-order mutual information (kNN, MINE); R kernel effective rank (RBF, median heuristic); M MDL (autoencoder).

Dataset	H	d_{pca}	$d_{2\text{nn}}$	K	T	T^*	I_2	I_h	R	M
adv-easy-looking-hard	12.03	1.93	10.93	0.97	955	0.06	0.01	0.02	4.58	80.31
adv-encoded-random	132	1.67	5.82	0.53	4.2e4	2.2e-3	0.01	0.07	4.16	473
adv-hard-looking-easy	-12.79	1.99	2.64	0.96	79.27	0.04	2.04	5.17	3.65	53.24
adv-quantum-trivial	-11.42	3.00	2.91	0.96	61.73	0.43	0.59	3.13	3.56	54.29
real-breast-cancer	101	1.04	4.91	0.53	2.1e4	8.0e-4	0.28	1.17	3.71	429
real-credit-default	225	2.13	7.16	0.20	1.8e7	2.6e-3	0.29	1.10	4.96	220
real-diabetes	30.30	1.27	6.50	0.24	9456	1.1e-3	0.07	0.34	4.59	83.69
real-digits-3v5	157	7.30	12.70	0.07	5728	0.11	0.04	0.26	7.21	1837
real-fashion-2v6	4036	5.90	20.17	0.13	6.1e5	0.04	0.17	0.68	9.70	4.3e4
real-higgs	31.11	15.90	18.21	0.56	1908	0.16	0.02	0.10	9.42	301
real-ionosphere	-3.32	7.78	11.80	0.43	422	6.1e-3	0.45	1.02	6.23	685
real-iris	1.68	1.43	3.70	0.22	28.99	2.7e-3	0.36	1.27	4.18	59.98
real-sonar	-41.57	5.93	9.03	0.37	132	0.03	0.07	0.47	6.77	2582
real-wine	46.25	1.00	3.10	0.31	2737	1.2e-5	0.12	0.75	2.75	269
synth-2sphere	0.41	2.99	2.89	0.96	59.79	0.42	0.40	0.96	3.56	21.68
synth-3sphere-pc	1.62	3.99	4.37	0.96	133	0.49	0.13	0.56	4.18	38.91
synth-anisotropic-gmm	6.55	4.46	10.17	0.97	487	6.4e-4	0.79	2.12	4.34	66.96
synth-concentric-rings	-13.99	2.00	2.86	0.96	52.43	0.07	1.61	4.90	3.45	54.15
synth-ea-spin-glass	-14.14	7.97	7.64	0.90	823	5.48	8.7e-3	4.1e-3	5.94	79.38
synth-klein-bottle	-0.15	3.99	2.91	0.96	91.66	0.66	0.37	0.62	4.16	39.03
synth-parity-2	-14.52	7.97	7.22	0.90	811	5.98	0.01	3.9e-3	5.95	80.45
synth-parity-3	-14.98	7.97	7.61	0.90	955	5.12	6.7e-3	2.2e-3	5.94	80.21
synth-parity-4	-14.51	7.98	7.93	0.90	894	6.07	0.01	2.1e-3	5.95	80.18
synth-parity-5	-14.87	7.97	7.78	0.90	897	5.41	8.2e-3	3.4e-3	5.94	79.27
synth-prng	6.73	7.96	9.99	0.95	555	0.32	7.5e-3	0.02	6.40	80.33
synth-sk-spin-glass	-14.21	7.97	7.88	0.90	878	5.28	5.5e-3	3.3e-3	5.94	79.86
synth-torus	2.58	2.22	3.04	0.96	128	0.22	0.30	0.16	3.53	21.65

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Table 16: Part I’s four complexity pillars and the composite C_{data} (Eq. (1), equal weights) for each classical-track dataset. C_{top} is the topological component as deployed for the pre-specified tests; C_{top}^* and C_{data}^* use the corrected estimator of Appendix A. Provided so the Part I quantities can be reproduced and compared; Part II does not use the composite (Sec. 3.1, Table 1). *Exploratory.*

Dataset	S	I_{corr}	K	C_{top}	C_{top}^*	C_{data}	C_{data}^*
adv-easy-looking-hard	12.03	0.03	0.97	955	0.06	968	13.08
adv-encoded-random	132	0.09	0.53	4.2e4	2.2e-3	4.2e4	132
adv-hard-looking-easy	-12.79	7.21	0.96	79.27	0.04	74.66	-4.57
adv-quantum-trivial	-11.42	3.72	0.96	61.73	0.43	55.00	-6.31
real-breast-cancer	101	1.45	0.53	2.1e4	8.0e-4	2.1e4	103
real-credit-default	225	1.40	0.20	1.8e7	2.6e-3	1.8e7	226
real-diabetes	30.30	0.41	0.24	9456	1.1e-3	9487	30.94
real-digits-3v5	157	0.31	0.07	5728	0.11	5885	157
real-fashion-2v6	4036	0.85	0.13	6.1e5	0.04	6.1e5	4037
real-higgs	31.11	0.13	0.56	1908	0.16	1940	31.96
real-ionosphere	-3.32	1.47	0.43	422	6.1e-3	420	-1.42
real-iris	1.68	1.63	0.22	28.99	2.7e-3	32.53	3.54
real-sonar	-41.57	0.54	0.37	132	0.03	91.79	-40.62
real-wine	46.25	0.88	0.31	2737	1.2e-5	2784	47.44
synth-2sphere	0.41	1.37	0.96	59.79	0.42	62.52	3.15
synth-3sphere-pc	1.62	0.69	0.96	133	0.49	136	3.76
synth-anisotropic-gmm	6.55	2.91	0.97	487	6.4e-4	497	10.43
synth-concentric-rings	-13.99	6.51	0.96	52.43	0.07	45.92	-6.44
synth-ea-spin-glass	-14.14	0.01	0.90	823	5.48	810	-7.74
synth-klein-bottle	-0.15	0.99	0.96	91.66	0.66	93.46	2.46
synth-parity-2	-14.52	0.01	0.90	811	5.98	797	-7.62
synth-parity-3	-14.98	9.0e-3	0.90	955	5.12	940	-8.96
synth-parity-4	-14.51	0.01	0.90	894	6.07	880	-7.53
synth-parity-5	-14.87	0.01	0.90	897	5.41	883	-8.55
synth-prng	6.73	0.02	0.95	555	0.32	563	8.02
synth-sk-spin-glass	-14.21	8.8e-3	0.90	878	5.28	865	-8.01
synth-torus	2.58	0.46	0.96	128	0.22	132	4.23

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